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(54) **COMPOSITION COMPRISING A
PARTICULAR OXIDATION DYE
PRECURSOR AND A PARTICULAR
CARBOXYLIC ACID**

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,589,578 A 6/1971 Kamal et al.
4,003,699 A 1/1977 Rose et al.
4,013,307 A 3/1977 Dowd et al.
4,137,180 A 1/1979 Naik et al.
RE30,199 E 1/1980 Rose et al.

4,823,985 A 4/1989 Grollier et al.
4,874,554 A 10/1989 Lange et al.
5,061,289 A 10/1991 Clausen et al.
5,380,340 A 1/1995 Neunhoeffer et al.
5,534,267 A 7/1996 Neunhoeffer et al.
5,663,366 A 9/1997 Neunhoeffer et al.
5,766,576 A 6/1998 Löwe et al.
6,099,592 A 8/2000 Vidal et al.
6,284,003 B1 9/2001 Rose et al.
6,730,789 B1 5/2004 Birault et al.
11,213,471 B2 1/2022 Nicou et al.
2013/0081647 A1 4/2013 Vohra et al.
2019/0224088 A1* 7/2019 Gross A61K 8/342
2020/0163851 A1* 5/2020 Nicou A61K 8/22

FOREIGN PATENT DOCUMENTS

DE 2359399 A1 6/1975
DE 3843892 A1 6/1990
DE 4133957 A1 4/1993
DE 19543988 A1 5/1997
EP 0770375 A1 5/1997
EP 3295923 A1* 3/2018 A61Q 5/10
FR 2586913 A1 3/1987
FR 2733749 A1 11/1996
FR 2801308 A1 5/2001
FR 2886136 A1 12/2006
FR 3045379 A1 6/2017
GB 1026978 A 4/1966
GB 1153196 A 5/1969
IT 201900005622 A1 10/2020
JP 02-019576 A 1/1990
JP 05-163124 A 6/1993
JP 2017160198 A* 9/2017 A61Q 5/06
WO 94/08969 A1 4/1994
WO 94/08970 A1 4/1994
WO 96/15765 A1 5/1996
WO 2022/129377 A1 6/2022

OTHER PUBLICATIONS

International Search Report and Written Opinion for counterpart
Application No. PCT/EP2021/086263, dated Apr. 21, 2022.
International Search Report and Written Opinion for counterpart
Application No. PCT/EP2021/086267, dated May 3, 2022.
Grevalcuore et al., "Hair Care Composition," Research Disclosure,
Kenneth Mason Publications, Hampshire, UK, GB, vol. 672, No.
75, Apr. 1, 2020, (4 pages).

(Continued)

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(57) **ABSTRACT**

The invention relates to a composition comprising at least
one oxidation dye coupler chosen from 6-hydroxy benzo-
morpholine of formula (I), hydroxyethyl-3,4-methy lenedi-
oxyaniline of formula (II). 2-amino-5-ethylphenol of for-
mula (III), one of their addition salts, their solvates and/or
the solvates of their salts, and at least one compound chosen
from N,N-dicarboxymethyl glutamic acid, one of its salts
and mixtures thereof. The invention also relates to a process
for dyeing keratin fibres, preferably the hair, comprising the
application of the composition to said keratin fibres.

20 Claims, No Drawings

(56)

References Cited

OTHER PUBLICATIONS

Porter, M.R., "Handbook of Surfactants," published by Blackie & Son (Glasgow and London), 1991, pp. 116-178.

Todd, Charles, et al., "Volatile Silicone Fluids for Cosmetic Formulations," *Cosmetics and Toiletries*, vol. 91, Jan. 1976, pp. 29-32.

* cited by examiner

1

**COMPOSITION COMPRISING A
PARTICULAR OXIDATION DYE
PRECURSOR AND A PARTICULAR
CARBOXYLIC ACID**

**CROSS REFERENCE TO RELATED
APPLICATIONS**

This is a National Stage Application of PCT/EP2021/086267, filed internationally on Dec. 16, 2021, which claims priority to French Application No. 2013469, filed on Dec. 17, 2020, both of which are incorporated by reference herein in their entireties.

The invention relates to a composition comprising the combination of a particular oxidation dye precursor and a particular carboxylic acid.

The invention also relates to a process for dyeing keratin fibres, in particular human keratin fibres such as hair, using this composition.

Finally, the invention relates to the use of such a composition for dyeing keratin fibres, and in particular human keratin fibres such as hair.

Many people have sought for a long time to modify the colour of their hair, and in particular to mask their grey hair.

It is known practice to dye keratin fibres, in particular human keratin fibres such as the hair, to obtain permanent colourings with dyeing compositions containing oxidation dye precursors, in particular oxidation bases, such as ortho- or para-phenylenediamines, ortho- or para-aminophenols, or heterocyclic compounds such as pyrazoles, pyrazolinones or pyrazolo-pyridines. These oxidation bases are colourless or weakly coloured compounds which, when combined with oxidizing products, can give rise to coloured compounds via a process of oxidative condensation.

It is also possible to vary the shades obtained with these oxidation bases by combining them with couplers or colour modifiers. The variety of molecules used as oxidation bases and couplers allows a wide range of colours to be obtained. However, the use of these dyeing compositions may have a certain number of drawbacks.

Specifically, after application to the keratin fibres, the dyeing power obtained may not be entirely satisfactory, or may even be weak, and lead to a restricted range of colours.

The colourings may also be insufficiently persistent with respect to external agents such as light, shampoo or perspiration, and may also be too selective, i.e. the difference in colouring is too great along the same keratin fibre that is differently sensitized between its end and its root.

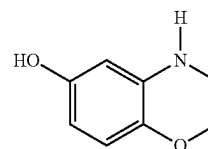
There is a real need to provide a composition for dyeing keratin fibres, in particular human keratin fibres such as the hair, which does not have the drawbacks mentioned above, i.e. which is capable of resulting in a colouring having an intense colour, with an improved fastness and also good coverage of the grey hair and good selectivity and which is capable of resulting in good dyeing performance levels, even after a period of storage.

These aims and others are achieved by the present invention, one subject of which is thus a composition, in particular for dyeing keratin fibres, comprising:

at least one oxidation coupler chosen from:

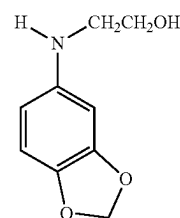
- i) 6-hydroxybenzomorpholine of formula (I), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof:

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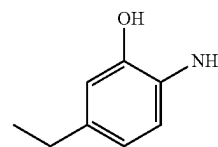
(I)

- ii) hydroxyethyl-3,4-methylenedioxyaniline of formula (II), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof:



(II)

- iii) 2-amino-5-ethylphenol of formula (III), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof:



(III)

and

at least one compound chosen from N,N-dicarboxymethyl glutamic acid, one of its salts, and mixtures thereof.

Another subject of the present invention is a process for dyeing keratin fibres in which the composition according to the invention is applied to said fibres.

According to a preferred embodiment, the composition according to the invention is a composition for dyeing keratin fibres, in particular the hair.

The composition according to the invention may especially lead to chromatic, powerful, intense and sparingly selective colourings, i.e. to colourings that are uniform along the length of the fibre. It also allows various shades to be achieved in a very wide range of colours. Furthermore, it enables a good colour build-up.

This composition also gives particularly good coverage of depigmented keratin fibres, such as grey hair.

A subject of the invention is also a kit comprising, in a first compartment, a composition as defined above and, in a second compartment, an oxidizing composition comprising at least one chemical oxidizing agent.

According to the invention, the term "chemical oxidizing agent" is intended to mean an oxidizing agent other than atmospheric oxygen.

Other subjects, characteristics, aspects and advantages of the invention will become even more clearly apparent on reading the description and the examples which follow.

In that which follows, and unless otherwise indicated, the limits of a range of values are included in this range, in particular in the expressions "of between" and "ranging from . . . to . . .".

3

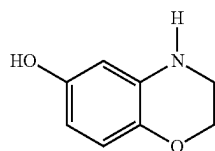
Moreover, the expression “at least one” used in the present description is equivalent to the expression “one or more”.

The composition according to the invention comprises at least one oxidation dye precursor.

Oxidation Couplers

The composition according to the invention comprises at least one particular oxidation coupler chosen from the compounds of formulae (I), (II) and (III), their addition salts, their solvates and/or the solvates of their salts, and mixtures thereof.

The coupler may be chosen from 6-hydroxybenzomorpholine of formula (I), one of its addition salts, its solvates and/or the solvates of its salts:

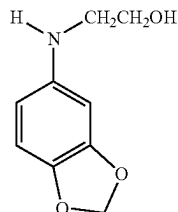


The addition salts of the compound of formula (I) present in the composition according to the invention are chosen especially from the addition salts with an acid, such as the hydrochlorides, hydrobromides, sulfates, citrates, succinates, tartrates, lactates, tosylates, benzenesulfonates, phosphates and acetates, and the addition salts with a base, such as sodium hydroxide, potassium hydroxide, aqueous ammonia, amines or alkanolamines.

Moreover, the solvates of the compound of formula (I) more particularly represent the hydrates of said compound and/or the combination of said compound with a linear or branched C₁ to C₄ alcohol such as methanol, ethanol, isopropanol or n-propanol. Preferably, the solvates are hydrates.

When it is present, the total content of coupler chosen from 6-hydroxybenzomorpholine of formula (I), one of its addition salts, its solvates and/or the solvates of its salts preferably ranges from 0.001% to 20% by weight, more preferably from 0.005% to 15% by weight, more preferentially from 0.01% to 10% by weight, better still from 0.05% to 5%, even better still from 0.1% to 3% by weight, relative to the total weight of the composition.

The coupler may be chosen from hydroxyethyl-3,4-methylenedioxyaniline of formula (II), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof:



The addition salts of the compounds of formula (II) are chosen, especially, from the addition salts with an acid, such as the hydrochlorides, hydrobromides, sulfates, citrates, succinates, tartrates, lactates, tosylates, benzenesulfonates, phosphates and acetates, and the addition salts with a base,

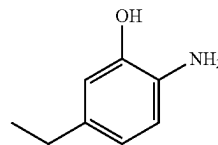
4

such as sodium hydroxide, potassium hydroxide, aqueous ammonia, amines or alkanolamines.

Moreover, the solvates of the compounds of formula (II) more particularly represent the hydrates of these compounds and/or the combination of these compounds with a linear or branched C₁ to C₄ alcohol such as methanol, ethanol, isopropanol or n-propanol. Preferably, the solvates are hydrates.

When it is present, the total content of the coupler chosen from hydroxyethyl-3,4-methylenedioxyaniline of formula (II), one of its addition salts, its solvates and/or the solvates of its salts preferably ranges from 0.001% to 20% by weight, more preferably from 0.005% to 15% by weight, more preferentially from 0.01% to 10% by weight, better still from 0.05% to 5%, even better still from 0.1% to 3% by weight, relative to the total weight of the composition.

The coupler may be chosen from 2-amino-5-ethylphenol of formula (III), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof:



The addition salts of the compounds of formula (III) are chosen, especially, from the addition salts with an acid, such as the hydrochlorides, hydrobromides, sulfates, citrates, succinates, tartrates, lactates, tosylates, benzenesulfonates, phosphates and acetates, and the addition salts with a base, such as sodium hydroxide, potassium hydroxide, aqueous ammonia, amines or alkanolamines.

Moreover, the solvates of the compounds of formula (III) more particularly represent the hydrates of these compounds and/or the combination of these compounds with a linear or branched C₁ to C₄ alcohol such as methanol, ethanol, isopropanol or n-propanol. Preferably, the solvates are hydrates.

When it is present, the total content of the coupler chosen from 2-amino-5-ethylphenol of formula (III), one of its addition salts, its solvates and/or the solvates of its salts preferably ranges from 0.001% to 20% by weight, more preferably from 0.005% to 15% by weight, more preferentially from 0.01% to 10% by weight, better still from 0.05% to 5%, even better still from 0.1% to 3% by weight, relative to the total weight of the composition.

The total content of coupler(s) chosen from 6-hydroxybenzomorpholine of formula (I), hydroxyethyl-3,4-methylenedioxyaniline of formula (II), 2-amino-5-ethylphenol of formula (III), one of their addition salts, their solvates and/or the solvates of their salts preferably ranges from 0.001% to 20% by weight, more preferably from 0.005% to 15% by weight, more preferentially from 0.01% to 10% by weight, better still from 0.05% to 5%, even better still from 0.1% to 3% by weight, relative to the total weight of the composition.

According to a preferred embodiment, the composition comprises at least one coupler chosen from 6-hydroxybenzomorpholine of formula (I), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof.

The composition according to the invention may optionally also comprise one or more additional couplers different from the compounds of formulae (I), (II) and (III) and from

their additional salts, their solvates and/or the solvates of their salts, advantageously chosen from those conventionally used in the dyeing of keratin fibres.

Among the additional couplers, mention may be made in particular of meta-phenylenediamines, meta-aminophenols, meta-diphenols, naphthalene-based coupling agents and heterocyclic coupling agents, other than the compounds of formulae (I), (II) and (III), and also the corresponding addition salts, the solvates and solvates of their salts.

Mention may be made, for example, of 1,3-dihydroxybenzene, 1,3-dihydroxy-2-methylbenzene, 4-chloro-1,3-dihydroxybenzene, 2,4-diamino-1-(β -hydroxyethoxy)benzene, 2-amino-4-(β -hydroxyethylamino)-1-methoxybenzene, 1,3-diaminobenzene, 1,3-bis(2,4-diaminophenoxy)propane, 3-ureidoaniline, 3-ureido-1-dimethylaminobenzene, sesamol, α -naphthol, 2-methyl-1-naphthol, 6-hydroxyindole, 4-hydroxyindole, 4-hydroxy-N-methylindole, 2-amino-3-hydroxypyridine, 3,5-diamino-2,6-dimethoxypyridine, 2,6-bis(β -hydroxyethylamino)toluene, 6-hydroxyindoline, 2,6-dihydroxy-4-methylpyridine, 1-H-3-methylpyrazol-5-one, 1-phenyl-3-methylpyrazol-5-one, 2,6-dimethylpyrazolo[1,5-b][1,2,4]triazole, 2,6-dimethyl[3,2-c][1,2,4]triazole and 6-methylpyrazolo[1,5-a]benzimidazole, 2-methyl-5-aminophenol, 5-N-(β -hydroxyethyl)amino-2-methylphenol, 3-aminophenol and 3-amino-2-chloro-6-methylphenol, the corresponding addition salts with an acid, the solvates and the solvates of the salts, and the corresponding mixtures.

These compounds may also be present in the form of solvates.

In a particular embodiment, the composition according to the invention is free from oxidation couplers chosen from resorcinol, 2-methylresorcinol, 4-chlororesorcinol, their addition salts, their solvates and the solvates of their salts.

In general, the addition salts of the additional couplers that may be used in the context of the invention are chosen in particular from the addition salts with an acid, such as the hydrochlorides, hydrobromides, sulfates, citrates, succinates, tartrates, lactates, tosylates, benzenesulfonates, phosphates and acetates, and the addition salts with a base, such as sodium hydroxide, potassium hydroxide, aqueous ammonia, amines or alkanolamines.

When the composition comprises one or more additional oxidation couplers, the total content of the additional coupler(s), different from the couplers of formulae (I), (II) and (III), from their salts, from their solvates and from the solvates of their salts that is (are) present, in the composition according to the invention preferably ranges from 0.001% to 20% by weight, more preferably from 0.005% to 15% by weight, more preferentially from 0.01% to 10% by weight, better still from 0.05% to 5%, even better still from 0.1% to 3% by weight, relative to the total weight of the composition.

Oxidation Bases

The composition according to the invention may also comprise at least one oxidation base.

Preferably, the composition according to the invention comprises at least one oxidation base.

By way of example, the oxidation bases are chosen from para-phenylenediamines, bis(phenyl)alkylenediamines, para-aminophenols, ortho-aminophenols, heterocyclic bases and the corresponding addition salts, the solvates and solvates of their salts.

Among the para-phenylenediamines that may be mentioned are, for example, para-phenylenediamine, para-toluenediamine, 2-chloro-para-phenylenediamine, 2,3-dimethyl-para-phenylenediamine, 2,6-dimethyl-para-pheny-

lenediamine, 2,6-diethyl-para-phenylenediamine, 2,5-dimethyl-para-phenylenediamine, N,N-dimethyl-para-phenylenediamine, N,N-diethyl-para-phenylenediamine, N,N-dipropyl-para-phenylenediamine, 4-amino-N,N-diethyl-3-methylaniline, N,N-bis(β -hydroxyethyl)-para-phenylenediamine, 4-N, N-bis(β -hydroxyethyl)amino-2-methylaniline, 4-N,N-bis(β -hydroxyethyl)amino-2-chloroaniline, 2- β -hydroxyethyl-para-phenylenediamine, 2-methoxymethyl-para-phenylenediamine, 2- γ -hydroxypropyl-para-phenylenediamine, 2-fluoro-para-phenylenediamine, 2-isopropyl-para-phenylenediamine, N-(β -hydroxypropyl)-para-phenylenediamine, 2-hydroxymethyl-para-phenylenediamine, N,N-dimethyl-3-methyl-para-phenylenediamine, N-ethyl-N-(β -hydroxyethyl)-para-phenylenediamine, N-(β , γ -dihydroxypropyl)-para-phenylenediamine, N-(4'-aminophenyl)-para-phenylenediamine, N-phenyl-para-phenylenediamine, 2- β -hydroxyethoxy-para-phenylenediamine, 2- β -acetylaminooxy-para-phenylenediamine, N-(β -methoxyethyl)-para-phenylenediamine, 4-aminophenylpyrrolidine, 2-thienyl-para-phenylenediamine, 2- β -hydroxyethylamino-5-aminotoluene and 3-hydroxy-1-(4'-aminophenyl)pyrrolidine, and the corresponding addition salts with an acid, the solvates and solvates of their salts.

Among the para-phenylenediamines mentioned above, particular preference is given to para-phenylenediamine, para-toluenediamine, 2-isopropyl-para-phenylenediamine, 2- β -hydroxyethyl-para-phenylenediamine, 2- β -hydroxyethoxy-para-phenylenediamine, 2-methoxymethyl-para-phenylenediamine, 2- γ -hydroxypropyl-para-phenylenediamine, 2,6-dimethyl-para-phenylenediamine, 2,6-diethyl-para-phenylenediamine, 2,3-dimethyl-para-phenylenediamine, N,N-bis(β -hydroxyethyl)-para-phenylenediamine, 2-chloro-para-phenylenediamine and 2- β -acetylaminooxy-para-phenylenediamine, and the corresponding addition salts with an acid, the solvates and solvates of their salts.

Among the bis(phenyl)alkylenediamines that may be mentioned are, for example, N,N'-bis(β -hydroxyethyl)-N,N'-bis(4'-aminophenyl)-1,3-diaminopropanol, N,N'-bis(β -hydroxyethyl)-N,N'-bis(4'-aminophenyl)ethylenediamine, N,N'-bis(4-aminophenyl)tetramethylenediamine, N,N'-bis(β -hydroxyethyl)-N,N'-bis(4-aminophenyl)tetramethylenediamine, N,N'-bis(4-methylaminophenyl)tetramethylenediamine, N,N'-bis(ethyl)-N,N'-bis(4'-amino-3'-methylphenyl)ethylenediamine and 1,8-bis(2,5-diaminophenoxy)-3,6-dioxaoctane, and the corresponding addition salts, the solvates and solvates of their salts.

Among the para-aminophenols that are mentioned are, for example, para-aminophenol, 4-amino-3-methylphenol, 4-amino-3-fluorophenol, 4-amino-3-chlorophenol, 4-amino-3-hydroxymethylphenol, 4-amino-2-methylphenol, 4-amino-2-hydroxymethylphenol, 4-amino-2-methoxymethylphenol, 4-amino-2-aminomethylphenol, 4-amino-2-(β -hydroxyethylaminomethyl)phenol and 4-amino-2-fluorophenol, and the corresponding addition salts with an acid, the solvates and solvates of their salts.

Among the ortho-aminophenols that may be mentioned are, for example, 2-aminophenol, 2-amino-5-methylphenol, 2-amino-6-methylphenol and 5-acetamido-2-aminophenol, and the corresponding addition salts, the solvates and solvates of their salts.

Among the heterocyclic bases that may be mentioned are, for example, pyridine, pyrimidine and pyrazole derivatives.

Among the pyridine derivatives that may be mentioned are the compounds described, for example, in patents GB 1 026 978 and GB 1 153 196, for example 2,5-diaminopyri-

dine, 2-(4-methoxyphenyl)amino-3-aminopyridine and 3,4-diaminopyridine, and the corresponding addition salts, the solvates and solvates of their salts.

Other pyridine oxidation bases that are useful in the present invention are the 3-aminopyrazolo[1,5-a]pyridine oxidation bases or the corresponding addition salts described, for example, in patent application FR 2 801 308. Examples that may be mentioned include pyrazolo[1,5-a]pyrid-3-ylamine, 2-acetylaminopyrazolo[1,5-a]pyrid-3-ylamine, 2-(morpholin-4-yl)pyrazolo[1,5-a]pyrid-3-ylamine, 3-aminopyrazolo[1,5-a]pyridine-2-carboxylic acid, 2-methoxypyrazolo[1,5-a]pyrid-3-ylamine, (3-aminopyrazolo[1,5-a]pyrid-7-yl)methanol, 2-(3-aminopyrazolo[1,5-a]pyrid-5-yl)ethanol, 2-(3-aminopyrazolo[1,5-a]pyrid-7-yl)ethanol, (3-aminopyrazolo[1,5-a]pyrid-2-yl)methanol, 3,6-diaminopyrazolo[1,5-a]pyridine, 3,4-diaminopyrazolo[1,5-a]pyridine, pyrazolo[1,5-a]pyridine-3,7-diamine, 7-(morpholin-4-yl)pyrazolo[1,5-a]pyrid-3-ylamine, pyrazolo[1,5-a]pyridine-3,5-diamine, 5-(morpholin-4-yl)pyrazolo[1,5-a]pyrid-3-ylamine, 2-[(3-aminopyrazolo[1,5-a]pyrid-5-yl)(2-hydroxyethyl)amino]ethanol, 2-[(3-aminopyrazolo[1,5-a]pyrid-7-yl)(2-hydroxyethyl)amino]ethanol, 3-aminopyrazolo[1,5-a]pyridin-5-ol, 3-aminopyrazolo[1,5-a]pyridin-4-ol, 3-aminopyrazolo[1,5-a]pyridin-6-ol, 3-aminopyrazolo[1,5-a]pyridin-7-ol, 2-3-hydroxyethoxy-3-aminopyrazolo[1,5-a]pyridine and 2-(4-dimethylpiperazinium-1-yl)-3-aminopyrazolo[1,5-a]pyridine, and the corresponding addition salts, the solvates and solvates of their salts.

More particularly, the oxidation bases that are useful in the present invention are chosen from 3-aminopyrazolo[1,5-a]pyridines and preferably substituted on carbon atom 2 with:

- a) a $(\text{di})(\text{C}_1-\text{C}_6)$ (alkyl)amino group, said alkyl group possibly being substituted with at least one hydroxyl, amino or imidazolium group;
- b) an optionally cationic 5- to 7-membered heterocycloalkyl group comprising from 1 to 3 heteroatoms, optionally substituted with one or more (C_1-C_6) alkyl groups, such as a $\text{di}(\text{C}_1-\text{C}_4)$ alkylpiperazinium group; or
- c) a (C_1-C_6) alkoxy group optionally substituted with one or more hydroxyl groups, such as a β -hydroxyalkoxy group, and the corresponding addition salts, the solvates and solvates of their salts.

Among the pyrimidine derivatives which may be mentioned are the compounds described, for example, in patents DE 2359399, JP 88-169571, JP 05-63124 and EP 0770375 or patent application WO 96/15765, such as 2,4,5,6-tetraaminopyrimidine, 4-hydroxy-2,5,6-triaminopyrimidine, 2-hydroxy-4,5,6-triaminopyrimidine, 2,4-dihydroxy-5,6-diaminopyrimidine, 2,5,6-triaminopyrimidine and their addition salts and their tautomeric forms, when a tautomeric equilibrium exists.

Among the pyrazole derivatives that may be mentioned are the compounds described in patents DE 3843892 and DE 4133957 and patent applications WO 94/08969, WO 94/08970, FR-A-2 733 749 and DE 195 43 988, for instance 4,5-diamino-1-methylpyrazole, 4,5-diamino-1-(β -hydroxyethyl)pyrazole, 3,4-diaminopyrazole, 4,5-diamino-1-(4'-chlorobenzyl)pyrazole, 4,5-diamino-1,3-dimethylpyrazole, 4,5-diamino-3-methyl-1-phenylpyrazole, 4,5-diamino-1-methyl-3-phenylpyrazole, 4-amino-1,3-dimethyl-5-hydrazinopyrazole, 1-benzyl-4,5-diamino-3-methylpyrazole, 4,5-diamino-3-tert-butyl-1-methylpyrazole, 4,5-diamino-1-tert-butyl-3-methylpyrazole, 4,5-diamino-1-(β -hydroxyethyl)-3-methylpyrazole, 4,5-diamino-1-ethyl-3-methylpyrazole, 4,5-diamino-1-ethyl-3-(4'-methoxyphenyl)

pyrazole, 4,5-diamino-1-ethyl-3-hydroxymethylpyrazole, 4,5-diamino-3-hydroxymethyl-1-methylpyrazole, 4,5-diamino-3-hydroxymethyl-1-isopropylpyrazole, 4,5-diamino-3-methyl-1-isopropylpyrazole, 4-amino-5-(2'-aminoethyl)amino-1,3-dimethylpyrazole, 3,4,5-triaminopyrazole, 1-methyl-3,4,5-triaminopyrazole, 3,5-diamino-1-methyl-4-methylaminopyrazole and 3,5-diamino-4-(β -hydroxyethyl)amino-1-methylpyrazole, and the corresponding addition salts, the solvates and solvates of their salts. Use may also be made of 4,5-diamino-1-(β -methoxyethyl)pyrazole.

A 4,5-diaminopyrazole will preferably be used and even more preferentially 4,5-diamino-1-(β -hydroxyethyl)pyrazole and/or a corresponding salt, the solvates and solvates of their salts.

The pyrazole derivatives which may also be mentioned comprise diamino-N,N-dihydropyrazolopyrazolones and in particular those described in patent application FR-A-2 886 136, such as the following compounds and the corresponding addition salts: 2,3-diamino-6,7-dihydro-1H,5H-pyrazolo[1,2-a]pyrazol-1-one, 2-amino-3-ethylamino-6,7-dihydro-1H,5H-pyrazolo[1,2-a]pyrazol-1-one, 2-amino-3-isopropylamino-6,7-dihydro-1H,5H-pyrazolo[1,2-a]pyrazol-1-one, 2-amino-3-(pyrrolidin-1-yl)-6,7-dihydro-1H,5H-pyrazolo[1,2-a]pyrazol-1-one, 4,5-diamino-1,2-dimethyl-1,2-dihydropyrazol-3-one, 4,5-diamino-1,2-diethyl-1,2-dihydropyrazol-3-one, 4,5-diamino-1,2-di(2-hydroxyethyl)-1,2-dihydropyrazol-3-one, 2-amino-3-(2-hydroxyethyl)amino-6,7-dihydro-1H,5H-pyrazolo[1,2-a]pyrazol-1-one, 2-amino-3-dimethylamino-6,7-dihydro-1H,5H-pyrazolo[1,2-a]pyrazol-1-one, 2,3-diamino-5,6,7,8-tetrahydro-1H,6H-pyridazino[1,2-a]pyrazol-1-one, 4-amino-1,2-diethyl-5-(pyrrolidin-1-yl)-1,2-dihydropyrazol-3-one, 4-amino-5-(3-dimethylaminopyrrolidin-1-yl)-1,2-diethyl-1,2-dihydropyrazol-3-one and 2,3-diamino-6-hydroxy-6,7-dihydro-1H,5H-pyrazolo[1,2-a]pyrazol-1-one.

Use will preferably be made of 2,3-diamino-6,7-dihydro-1H,5H-pyrazolo[1,2-a]pyrazol-1-one and/or a corresponding salt, the solvates and solvates of their salts.

Use will preferably be made, as heterocyclic bases, of 4,5-diamino-1-(β -hydroxyethyl)pyrazole and/or 2,3-diamino-6,7-dihydro-1H,5H-pyrazolo[1,2-a]pyrazol-1-one and/or 2- β -hydroxyethoxy-3-aminopyrazolo[1,5-a]pyridine and/or a corresponding salt.

These compounds may also be present in the form of solvates. The addition salts of the oxidation bases which may be present in the composition according to the invention are chosen especially from the addition salts with an acid, such as the hydrochlorides, hydrobromides, sulfates, succinates, citrates, tartrates, lactates, tosylates, benzene-sulfonates, methanesulfonates, phosphates and acetates, and the addition salts with a base, such as sodium hydroxide, potassium hydroxide, aqueous ammonia, amines or alkanolamines.

Moreover, the solvates of the additional oxidation bases more particularly represent the hydrates of said oxidation bases and/or the combination of said oxidation bases with a linear or branched C_1 to C_4 alcohol such as methanol, ethanol, isopropanol or n-propanol. Preferably, the solvates are hydrates.

Preferably, the oxidation base(s) is (are) chosen from para-phenylenediamines, bis(phenyl)alkylenediamines, para-aminophenols, ortho-aminophenols, heterocyclic bases, and the corresponding addition salts, and mixtures thereof; more preferentially from 2-methoxymethyl-para-phenylenediamine, 2- β -hydroxyethyl-para-phenylenediamine, 2- γ -hydroxypropyl-para-phenylenediamine, N-(β -

hydroxypropyl)-para-phenylenediamine, their addition salts, their solvates, the solvates of their salts and mixtures thereof.

In a particular embodiment, the composition according to the invention is free from oxidation bases chosen from para-phenylenediamine, para-toluenediamine, their addition salts, their solvates and the solvates of their salts.

When it (they) is (are) present, the oxidation base(s) is (are) present in a total content ranging from 0.001% to 20% by weight, more preferably from 0.005% to 15% by weight, more preferentially from 0.01% to 10% by weight, better still from 0.05% to 5%, even better still from 0.1% to 3%, relative to the weight of the composition.

According to a preferred embodiment, the oxidation base(s) chosen from 2-methoxymethyl-para-phenylenediamine, 2- β -hydroxyethyl-para-phenylenediamine, 2- γ -hydroxypropyl-para-phenylenediamine, one of their addition salts, their solvates, the solvates of their salts and mixtures thereof is (are) present in a total content ranging from 0.001% to 20% by weight, more preferably from 0.005% to 15% by weight, more preferentially from 0.01% to 10% by weight, better still from 0.05% to 5%, even better still from 0.1% to 3% by weight, relative to the weight of the composition.

Advantageously, the weight ratio between the total content of the oxidation base(s) and the total content of the coupler(s) chosen from 6-hydroxybenzomorpholine of formula (I), hydroxyethyl-3,4-methylenedioxyaniline of formula (II), 2-amino-5-ethylphenol of formula (III), one of their addition salts, their solvates and/or the solvates of their salts ranges from 0.1 to 10, preferably from 0.5 to 5.

Advantageously, the weight ratio between the total content of the oxidation base(s) and the total content of the coupler(s) ranges from 0.1 to 10, preferably from 0.3 to 3.

N,N-dicarboxymethyl Glutamic Acid

The composition according to the invention comprises at least one compound chosen from N,N-dicarboxymethyl glutamic acid, one of its salts and mixtures thereof.

The salts are in particular alkali metal, alkaline-earth metal, ammonium and substituted ammonium salts.

Among the salts of these compounds, the alkali metal salts and especially the sodium or potassium salts are preferred.

The composition according to the invention preferably comprises tetrasodium glutamate diacetate (GLDA). Use will be made, for example, of Dissolvine GL38 or 45S from Akzo Nobel.

Preferably, the total content of the compound(s) chosen from N,N-dicarboxymethyl glutamic acid, its salts and mixtures thereof, ranges preferably from 0.001 to 15% by weight, more preferably from 0.005 to 10% by weight, preferentially from 0.01 to 8% by weight, better still from 0.05 to 5% by weight, even better still from 0.075 to 2% by weight, relative to the total weight of the composition.

Fatty Substances

The composition according to the invention may comprise one or more fatty substances.

The term "fatty substance" is understood to mean an organic compound that is insoluble in water at 25° C. and at atmospheric pressure (1.013×10^5 Pa) (solubility of less than 5% by weight, and preferably less than 1% by weight, even more preferentially less than 0.1% by weight). They bear in their structure at least one hydrocarbon-based chain including at least 6 carbon atoms and/or a sequence of at least two siloxane groups. In addition, the fatty substances are generally soluble in organic solvents under the same temperature and pressure conditions, for instance chloroform,

dichloromethane, carbon tetrachloride, ethanol, benzene, toluene, tetrahydrofuran (THF), liquid petroleum jelly or decamethylcyclopentasiloxane.

Advantageously, the fatty substances that may be used in the present invention are neither (poly)oxyalkylenated nor (poly)glycerolated.

Preferably, the fatty substances that are of use according to the invention are non-silicone.

The term "non-silicone fatty substance" is intended to mean a fatty substance not containing any Si—O bonds and the term "silicone fatty substance" is intended to mean a fatty substance containing at least one Si—O bond.

Useful fatty substances according to the invention may be liquid fatty substances (or oils) and/or solid fatty substances.

A liquid fatty substance is understood to be a fatty substance having a melting point of less than or equal to 25° C. at atmospheric pressure (1.013×10^5 Pa). A solid fatty substance is understood to be a fatty substance having a melting point of greater than 25° C. at atmospheric pressure (1.013×10^5 Pa).

For the purposes of the present invention, the melting point corresponds to the temperature of the most endothermic peak observed on thermal analysis (differential scanning calorimetry or DSC) as described in the standard ISO 11357-3; 1999. The melting point may be measured using a differential scanning calorimeter (DSC), for example the calorimeter sold under the name "MDSC 2920" by the company TA Instruments. In the present application, all the melting points are determined at atmospheric pressure (1.013×10^5 Pa).

More particularly, the liquid fatty substance(s) according to the invention is (are) chosen from C_6 to C_{16} liquid hydrocarbons, liquid hydrocarbons comprising more than 16 carbon atoms, non-silicone oils of animal origin, oils of triglyceride type of plant or synthetic origin, fluoro oils, liquid fatty alcohols, liquid esters of fatty acid and/or of fatty alcohol other than triglycerides, and silicone oils, and mixtures thereof.

It is recalled that the fatty alcohols, esters and acids more particularly contain at least one saturated or unsaturated, linear or branched hydrocarbon-based group, comprising from 6 to 40 and better still from 8 to 30 carbon atoms, which is optionally substituted, in particular with one or more hydroxyl groups (in particular 1 to 4). If they are unsaturated, these compounds may comprise one to three conjugated or unconjugated carbon-carbon double bonds.

As regards the C_6 to C_{16} liquid hydrocarbons, the latter may be linear, branched, or optionally cyclic, and are preferably chosen from alkanes. Examples that may be mentioned include hexane, cyclohexane, undecane, dodecane, isododecane, tridecane or isoparaffins, such as isohexadecane or isodecane, and mixtures thereof.

The liquid hydrocarbons comprising more than 16 carbon atoms may be linear or branched, of mineral or synthetic origin, and are preferably chosen from liquid paraffin or liquid petroleum jelly (INCI name mineral oil or paraffinum liquidum), polydecenes, hydrogenated polyisobutene such as Parleam®, and mixtures thereof.

A hydrocarbon-based oil of animal origin that may be mentioned is perhydrosqualene.

The triglyceride oils of plant or synthetic origin are preferably chosen from liquid triglycerides of fatty acids comprising from 6 to 30 carbon atoms, for instance heptanoic or octanoic acid triglycerides, or alternatively, for example, sunflower oil, corn oil, soybean oil, marrow oil, grapeseed oil, sesame seed oil, hazelnut oil, apricot oil, macadamia oil, arara oil, sunflower oil, castor oil, avocado

oil, caprylic/capric acid triglycerides, for instance those sold by the company Stéarinerie Dubois or those sold under the names Miglyol® 810, 812 and 818 by the company Dynamit Nobel, jojoba oil and shea butter oil, and mixtures thereof.

As regards the fluoro oils, they may be chosen from perfluoromethylcyclopentane and perfluoro-1,3-dimethylcyclohexane, sold under the names Flutec® PC1 and Flutec® PC3 by the company BNFL Fluorochemicals; perfluoro-1,2-dimethylcyclobutane; perfluoroalkanes such as dodecafluoropentane and tetradecafluorohexane, sold under the names PF 5050® and PF 5060® by the company 3M, or else bromoperfluorooctyl sold under the name Foralkyl® by the company Atochem; nonafluoromethoxybutane and nonafluoroethoxyisobutane; perfluoromorpholine derivatives, such as 4-trifluoromethylperfluoromorpholine sold under the name PF 5052® by the company 3M.

The liquid fatty alcohols that are suitable for use in the invention are more particularly chosen from linear or branched, saturated or unsaturated alcohols, preferably unsaturated or branched alcohols, comprising from 6 to 40 carbon atoms, preferably from 8 to 30 carbon atoms. Examples that may be mentioned include octyldodecanol, 2-butyloctanol, 2-hexyldecanol, 2-undecylpentadecanol, isostearyl alcohol, oleyl alcohol, linolenyl alcohol, ricinoleyl alcohol, undecylenyl alcohol and linoleyl alcohol, and mixtures thereof.

As regards the liquid esters of fatty acids and/or of fatty alcohols, other than the triglycerides mentioned previously, mention may be made especially of esters of saturated or unsaturated, linear C1 to C26 or branched C3 to C26 aliphatic mono- or polyacids and of saturated or unsaturated, linear C1 to C26 or branched C3 to C26 aliphatic mono- or polyalcohols, the total carbon number of the esters being greater than or equal to 6, more advantageously greater than or equal to 10.

Preferably, for the esters of monoalcohols, at least one from among the alcohol and the acid from which the esters of the invention are derived is branched.

Among the monoesters, mention may be made of dihydroabietyl behenate; octyldodecyl behenate; isocetyl behenate; isostearyl lactate; lauryl lactate; linoleyl lactate; oleyl lactate; isostearyl octanoate; isocetyl octanoate; octyl octanoate; decyl oleate; isocetyl isostearate; isocetyl laurate; isocetyl stearate; isodecyl octanoate; isodecyl oleate; isononyl isononanoate; isostearyl palmitate; methyl acetyl ricinoleate; octyl isononanoate; 2-ethylhexyl isononanoate; octyldodecyl erucate; oleyl erucate; ethyl palmitate, isopropyl palmitate, such as 2-ethylhexyl palmitate, 2-octyldecyl palmitate; alkyl myristates such as isopropyl myristate; isobutyl stearate; 2-hexyldecyl laurate, and mixtures thereof.

Preferably, among the monoesters of monoacids and of monoalcohols, use will be made of ethyl palmitate and isopropyl palmitate, alkyl myristates such as isopropyl myristate or ethyl myristate, isocetyl stearate, 2-ethylhexyl isononanoate, isodecyl neopentanoate and isostearyl neopentanoate, and mixtures thereof.

Still within the context of this variant, esters of C₄ to C₂₂ dicarboxylic or tricarboxylic acids and of C₁ to C₂₂ alcohols and esters of mono-, di- or tricarboxylic acids and of C₂ to C₂₆ di-, tri-, tetra- or pentahydroxy alcohols may also be used.

Mention may especially be made of: diethyl sebacate; diisopropyl sebacate; diisopropyl adipate; di-n-propyl adipate; dioctyl adipate; diisostearyl adipate; dioctyl maleate; glyceryl undecylenate; octyldodecyl stearoyl stearate; pentaerythrityl monoricinoleate; pentaerythrityl tetraisononanoate; pentaerythrityl tetrapelargonate; pentaerythrityl tet-

raisostearate; pentaerythrityl tetraoctanoate; propylene glycol dicaprylate; propylene glycol dicaprate; tridecyl erucate; triisopropyl citrate; triisostearyl citrate; glyceryl trilactate; glyceryl trioctanoate; trioctyldodecyl citrate; trioleyl citrate; propylene glycol dioctanoate; neopentyl glycol diheptanoate; diethylene glycol diisononanoate; polyethylene glycol distearates, and mixtures thereof.

The composition may also comprise, as fatty ester, sugar esters and diesters of C₆ to C₃₀, preferably of C₁₂ to C₂₂, fatty acids. It is recalled that the term "sugar" refers to oxygen-bearing hydrocarbon-based compounds bearing several alcohol functions, with or without aldehyde or ketone functions, and which include at least 4 carbon atoms. These sugars may be monosaccharides, oligosaccharides or polysaccharides.

Examples of suitable sugars that may be mentioned include sucrose, glucose, galactose, ribose, fucose, maltose, fructose, mannose, arabinose, xylose and lactose, and derivatives thereof, especially alkyl derivatives, such as methyl derivatives, for instance methylglucose.

The sugar esters of fatty acids may be chosen especially from the group comprising the esters or mixtures of esters of sugars described above and of linear or branched, saturated or unsaturated C6 to C30 and preferably C12 to C22 fatty acids. If they are unsaturated, these compounds may comprise one to three conjugated or unconjugated carbon-carbon double bonds.

The esters according to this variant may also be chosen from mono-, di-, tri- and tetraesters, polyesters, and mixtures thereof.

These esters may be, for example, oleates, laurates, palmitates, myristates, behenates, cocoates, stearates, linoleates, linolenates, caprates, arachidonates or mixtures thereof, for instance especially the mixed oleopalmitate, oleostearate and palmitostearate esters.

More particularly, use is made of monoesters and diesters and especially sucrose, glucose or methylglucose mono- or dioleates, -stearates, -behenates, -oleopalmitates, -linoleates, -linolenates and -oleostearates, and mixtures thereof.

An example that may be mentioned is the product sold under the name Glucate® DO by the company Amerchol, which is a methylglucose dioleate.

Preferably, use will be made of a liquid ester of a monoacid and of a monoalcohol.

The silicone oils that may be used in the composition according to the present invention may be volatile or non-volatile, cyclic, linear or branched silicone oils, which are unmodified or modified with organic groups, and preferably have a viscosity from 5×10^{-6} to $2.5 \text{ m}^2/\text{s}$ at 25° C., and preferably 1×10^{-5} to $1 \text{ m}^2/\text{s}$.

Preferably, the silicone oils are chosen from polydialkylsiloxanes, especially polydimethylsiloxanes (PDMS), and liquid polyorganosiloxanes including at least one aryl group.

These silicone oils may also be organomodified. The organomodified silicone oils that may be used in accordance with the invention are preferably liquid silicones as defined above and including in their structure one or more organofunctional groups attached via a hydrocarbon-based group, chosen, for example, from amine groups and alkoxy groups.

Organopolysiloxanes are defined in greater detail in Walter Noll's "Chemistry and Technology of Silicones" (1968), Academic Press. They may be volatile or non-volatile.

When they are volatile, the silicone oils are more particularly chosen from those with a boiling point of between 60° C. and 260° C., and even more particularly from:

(i) cyclic polydialkylsiloxanes including from 3 to 7 and preferably from 4 to 5 silicon atoms. These are, for

example, octamethylcyclotetrasiloxane sold notably under the name Volatile Silicone® 7207 by Union Carbide or Silbione® 70045 V2 by Rhodia, decamethylcyclopentasiloxane sold under the name Volatile Silicone® 7158 by Union Carbide, and Silbione® 70045 V5 by Rhodia, and mixtures thereof.

Mention may also be made of cyclocopolymers of the dimethylsiloxane/methylalkylsiloxane type, such as Volatile Silicone® FZ 3109 sold by the company Union Carbide.

Mention may also be made of mixtures of cyclic polydialkylsiloxanes with organosilicon compounds, such as the mixture of octamethylcyclotetrasiloxane and tetra(trimethylsilyl)pentaerythritol (50/50) and the mixture of octamethylcyclotetrasiloxane and oxy-1,1'-bis(2,2,2',3,3'-hexatrimethylsilyloxy)neopentane;

(ii) linear volatile polydialkylsiloxanes containing 2 to 9 silicon atoms and having a viscosity of less than or equal to 5×10^{-6} m²/s at 25° C. An example is decamethyltetrasiloxane sold in particular under the name SH 200 by the company Toray Silicone. Silicones falling within this category are also described in the article published in *Cosmetics and Toiletries*, Vol. 91, January 76, pp. 27-32, Todd & Byers, "Volatile Silicone Fluids for Cosmetics".

Non-volatile polydialkylsiloxanes are preferably used.

These silicone oils are more particularly chosen from polydialkylsiloxanes, among which mention may be made mainly of polydimethylsiloxanes bearing trimethylsilyl end groups.

The viscosity of the silicones is measured at 25° C. according to the standard ASTM 445, Appendix C.

Among these polydialkylsiloxanes, mention may be made, in a nonlimiting manner, of the following commercial products:

the Silbione® oils of the 47 and 70 047 series or the Mirasil® oils sold by Rhodia, such as, for example, the oil 70 047 V 500,000;

the oils of the Mirasil® series sold by the company Rhodia;

the oils of the 200 series from the company Dow Corning, such as DC200 having a viscosity of 60,000 mm²/s; the Viscasil® oils from General Electric and certain oils of the SF series (SF 96, SF 18) from General Electric.

Mention may also be made of polydimethylsiloxanes bearing dimethylsilyl end groups, known under the name dimethiconol (CTFA), such as the oils of the 48 series from the company Rhodia.

The organomodified silicones that may be used in accordance with the invention are silicones as defined above and comprising in their structure one or more organofunctional groups attached via a hydrocarbon-based group.

As regards the liquid polyorganosiloxanes including at least one aryl group, they may especially be polydiphenylsiloxanes, and polyalkylarylsiloxanes functionalized with the organofunctional groups mentioned previously.

The polyalkylarylsiloxanes are particularly chosen from linear and/or branched polydimethyl/methylphenylsiloxanes and polydimethyl/diphenylsiloxanes with a viscosity ranging from 1×10^{-5} to 5×10^{-2} m²/s at 25° C.

Among these polyalkylarylsiloxanes, examples that may be mentioned include the products sold under the following names:

the Silbione® oils of the 70 641 series from Rhodia;

the oils of the Rhodorsil® 70 633 and 763 series from Rhodia;

the oil Dow Corning 556 Cosmetic Grade Fluid from Dow Corning;

the silicones of the PK series from Bayer, such as the product PK20;

the silicones of the PN and PH series from Bayer, such as the products PN1000 and PH1000;

certain oils of the SF series from General Electric, such as SF 1023, SF 1154, SF 1250 and SF 1265.

Among the organomodified silicones, mention may be made of polyorganosiloxanes including:

substituted or unsubstituted amine groups, such as the products sold under the names GP 4 Silicone Fluid and GP 7100 by the company Genesee or the products sold under the names Q2 8220 and Dow Corning 929 or 939 by the company Dow Corning. The substituted amine groups are in particular C1 to C4 aminoalkyl groups;

alkoxy groups,

hydroxyl groups.

The solid fatty substances according to the invention preferably have a viscosity of greater than 2 Pa·s, measured at 25° C. and at a shear rate of 1 s⁻¹.

The solid fatty substance(s) is (are) preferably chosen from solid fatty acids, solid fatty alcohols, solid esters of fatty acids and/or of fatty alcohols, waxes, ceramides and mixtures thereof.

The term "fatty acids" is intended to mean a long-chain carboxylic acid comprising from 6 to 40 carbon atoms, preferably from 8 to 30 carbon atoms. The solid fatty acids according to the invention preferentially comprise from 10 to 30 carbon atoms and better still from 14 to 22 carbon atoms. They may optionally be hydroxylated. These fatty acids are neither oxyalkylenated nor glycerolated.

The solid fatty acids that may be used in the present invention may especially be chosen from myristic acid, cetyl acid (palmitic acid), arachidic acid, stearic acid, lauric acid, behenic acid, 12-hydroxystearic acid, and mixtures thereof.

Particularly preferably, the solid fatty acid(s) is (are) chosen from lauric acid, myristic acid, cetyl acid (palmitic acid), stearic acid.

The term "fatty alcohol" is intended to mean a long-chain aliphatic alcohol comprising from 6 to 40 carbon atoms, preferably from 8 to 30 carbon atoms, and comprising at least one hydroxyl group OH. These fatty alcohols are neither oxyalkylenated nor glycerolated.

The solid fatty alcohols may be saturated or unsaturated, and linear or branched, and include from 8 to 40 carbon atoms, preferably from 10 to 30 carbon atoms. Preferably, the solid fatty alcohols have the structure R—OH with R denoting a linear alkyl group, optionally substituted with one or more hydroxyl groups, comprising from 8 to 40, preferentially from 10 to 30 carbon atoms, better still from 10 to 30, or even from 12 to 24 atoms, and even better still from 14 to 22 carbon atoms.

The solid fatty alcohols that may be used are preferably chosen from saturated or unsaturated, linear or branched, preferably linear and saturated, (mono)alcohols including from 8 to 40 carbon atoms, better still from 10 to 30, or even from 12 to 24 atoms, and even better still from 14 to 22 carbon atoms.

The solid fatty alcohols that may be used may be chosen, alone or as a mixture, from: myristyl alcohol (or 1-tetradecanol); cetyl alcohol (or 1-hexadecanol); stearyl alcohol (or 1-octadecanol); arachidyl alcohol (or 1-eicosanol); behenyl alcohol (or 1-docosanol); lignoceryl alcohol (or 1-tetracosanol); ceryl alcohol (or 1-hexacosanol); montanyl alcohol (or 1-octacosanol); myricyl alcohol (or 1-triacontanol).

Preferentially, the solid fatty alcohol is chosen from cetyl alcohol, stearyl alcohol, behenyl alcohol, myristyl alcohol,

arachidyl alcohol, and mixtures thereof, such as cetylstearyl alcohol or cetearyl alcohol. Particularly preferably, the solid fatty alcohol is cetylstearyl or cetearyl alcohol.

The solid esters of a fatty acid and/or of a fatty alcohol that may be used are preferably chosen from esters resulting from a C₉-C₂₆ carboxylic fatty acid and/or from a C₉-C₂₆ fatty alcohol.

Preferably, these solid fatty esters are esters of a linear or branched, saturated carboxylic acid including at least 10 carbon atoms, preferably from 10 to 30 carbon atoms and more particularly from 12 to 24 carbon atoms, and of a linear or branched, saturated monoalcohol including at least 10 carbon atoms, preferably from 10 to 30 carbon atoms and more particularly from 12 to 24 carbon atoms. The saturated carboxylic acids may be optionally hydroxylated, and are preferably monocarboxylic acids.

Esters of C₄-C₂₂ dicarboxylic or tricarboxylic acids and of C₁-C₂₂ alcohols and esters of mono-, di- or tricarboxylic acids and of C₂-C₂₆ di-, tri-, tetra- or pentahydroxylated alcohols may also be used.

Mention may especially be made of octyldodecyl behenate, isocetyl behenate, cetyl lactate, stearyl octanoate, octyl octanoate, cetyl octanoate, decyl oleate, hexyl stearate, octyl stearate, myristyl stearate, cetyl stearate, stearyl stearate, octyl pelargonate, cetyl myristate, myristyl myristate, stearyl myristate, diethyl sebacate, diisopropyl sebacate, diisopropyl adipate, di-n-propyl adipate, dioctyl adipate, dioctyl maleate, octyl palmitate, myristyl palmitate, cetyl palmitate, stearyl palmitate, and mixtures thereof.

Preferably, the solid esters of a fatty acid and/or of a fatty alcohol are chosen from C₉-C₂₆ alkyl palmitates, notably myristyl palmitate, cetyl palmitate or stearyl palmitate; C₉-C₂₆ alkyl myristates, such as cetyl myristate, stearyl myristate and myristyl myristate; C₉-C₂₆ alkyl stearates, in particular myristyl stearate, cetyl stearate and stearyl stearate; and mixtures thereof.

For the purposes of the present invention, a wax is a lipophilic compound, which is solid at 25° C. and atmospheric pressure, with a reversible solid/liquid change of state, having a melting point greater than about 40° C. and which may range up to 200° C., and having in the solid state an anisotropic crystal organisation. In general, the size of the wax crystals is such that the crystals diffract and/or scatter light, giving the composition that comprises them a more or less opaque cloudy appearance. By bringing the wax to its melting point, it is possible to make it miscible with oils and to form a microscopically homogeneous mixture, but on returning the temperature of the mixture to ambient temperature, recrystallization of the wax, which is microscopically and macroscopically detectable (opalescence), is obtained.

In particular, the waxes that are suitable for use in the invention may be chosen from waxes of animal, plant or mineral origin, non-silicone synthetic waxes, and mixtures thereof.

Mention may be made notably of hydrocarbon-based waxes, for instance beeswax, notably of biological origin, lanolin wax, and Chinese insect waxes; rice bran wax, carnauba wax, candelilla wax, ouricury wax, esparto grass wax, berry wax, shellac wax, Japan wax and sumach wax; montan wax, orange wax and lemon wax, microcrystalline waxes, paraffins and ozokerite; polyethylene waxes, the waxes obtained by Fischer-Tropsch synthesis and waxy copolymers, and also esters thereof.

Mention may also be made of C₂₀ to C₆₀ microcrystalline waxes, such as Microwax HW.

Mention may also be made of the MW 500 polyethylene wax sold under the reference Permalen 50-L polyethylene.

Mention may also be made of waxes obtained by catalytic hydrogenation of animal or plant oils containing linear or branched C₈ to C₃₂ fatty chains. Among these waxes, mention may notably be made of isomerized jojoba oil, such as the trans-isomerized partially hydrogenated jojoba oil, notably the product manufactured or sold by the company Desert Whale under the commercial reference Iso-Jojoba-50®, hydrogenated sunflower oil, hydrogenated castor oil, hydrogenated coconut oil, hydrogenated lanolin oil, and bis(1,1,1-trimethylolpropane) tetrastearate, notably the product sold under the name Hest 2T-4S® by the company Heterene.

The waxes obtained by hydrogenation of castor oil esterified with cetyl alcohol, such as those sold under the names Phytowax Castor 16L64® and 22L73® by the company Sophim, may also be used.

A wax that may also be used is a C₂₀ to C₄₀ alkyl (hydroxystearyl)stearate (the alkyl group comprising from 20 to 40 carbon atoms), alone or as a mixture. Such a wax is notably sold under the names "Kester Wax K 82 P®", "Hydroxypolyester K 82 P®" and "Kester Wax K 80 PR" by the company Koster Keunen.

It is also possible to use microwaxes in the compositions of the invention; mention may be made notably of carnauba microwaxes, such as the product sold under the name MicroCare 350® by the company Micro Powders, synthetic-wax microwaxes, such as the product sold under the name MicroEase 114S® by the company Micro Powders, microwaxes constituted of a mixture of carnauba wax and of polyethylene wax, such as the products sold under the names Micro Care 300® and 310® by the company Micro Powders, microwaxes constituted of a mixture of carnauba wax and of synthetic wax, such as the product sold under the name Micro Care 325® by the company Micro Powders, polyethylene microwaxes, such as the products sold under the names Micropoly 200®, 220®, 220L® and 250S® by the company Micro Powders, and polytetrafluoroethylene microwaxes, such as the products sold under the names Microslip 519® and 519 L® by the company Micro Powders.

The waxes are preferably chosen from mineral waxes, for instance paraffin, petroleum jelly, lignite or ozokerite wax; plant waxes, for instance cocoa butter or cork fibre or sugar cane waxes, olive tree wax, rice wax, hydrogenated jojoba wax, ouricury wax, carnauba wax, candelilla wax, esparto grass wax, or absolute waxes of flowers, such as the essential wax of blackcurrant blossom sold by the company Bertin (France); waxes of animal origin, for instance beeswaxes or modified beeswaxes (cera bellina), spermaceti, lanolin wax and lanolin derivatives; microcrystalline waxes; and mixtures thereof.

The ceramides, or ceramide analogues such as glycosceramides, that may be used in the compositions according to the invention are known; mention may in particular be made of ceramides of classes I, II, III and V according to the Dawning classification.

The ceramides or analogues thereof that may be used preferably correspond to the following formula: R³CH(OH)CH(CH₂OR²)(NHCOR¹), in which:

R¹ denotes a linear or branched, saturated or unsaturated alkyl group, derived from C₁₄-C₃₀ fatty acids, it being possible for this group to be substituted with a hydroxyl group in the alpha position, or a hydroxyl group in the omega position esterified with a saturated or unsaturated C₁₆-C₃₀ fatty acid;

17

R² denotes a hydrogen atom, a (glycosyl)_n group, a (galactosyl)_m group or a sulfogalactosyl group, in which n is an integer ranging from 1 to 4 and m is an integer ranging from 1 to 8;

R³ denotes a C₁₅-C₂₆ hydrocarbon-based group, saturated or unsaturated in the alpha position, it being possible for this group to be substituted with one or more C₁-C₁₄ alkyl groups; it being understood that in the case of natural ceramides or glycosceramides, R³ may also denote a C₁₅-C₂₆ alpha-hydroxyalkyl group, the hydroxyl group being optionally esterified with a C₁₆-C₃₀ alpha-hydroxy acid.

The ceramides that are more particularly preferred are the compounds for which R¹ denotes a saturated or unsaturated alkyl derived from C₁₆-C₂₂ fatty acids; R² denotes a hydrogen atom and R³ denotes a saturated linear C₁₅ group.

Preferentially, use is made of ceramides for which R¹ denotes a saturated or unsaturated alkyl group derived from C₁₄-C₃₀ fatty acids; R² denotes a galactosyl or sulfogalactosyl group; and R³ denotes a —CH=CH—(CH₂)₁₂—CH₃ group.

Use may also be made of the compounds for which R¹ denotes a saturated or unsaturated alkyl radical derived from C₁₂-C₂₂ fatty acids; R² denotes a galactosyl or sulfogalactosyl radical and R³ denotes a saturated or unsaturated C₁₂-C₂₂ hydrocarbon-based radical and preferably a —CH=CH—(CH₂)₁₂—CH₃ group.

As compounds that are particularly preferred, mention may also be made of 2-N-linoleoylaminoctadecane-1,3-diol; 2-N-oleoylaminoctadecane-1,3-diol; 2-N-palmitoylaminoctadecane-1,3-diol; 2-N-stearoylaminoctadecane-1,3-diol; 2-N-behenoylaminoctadecane-1,3-diol; 2-N-[2-hydroxypalmitoyl]aminoctadecane-1,3-diol; 2-N-stearoylaminoctadecane-1,3,4-triol and in particular N-stearoylphytylphosphingosine, 2-N-palmitoylaminoctadecane-1,3-diol, N-linoleoyldihydrosphingosine, N-oleoyldihydrosphingosine, N-palmitoyldihydrosphingosine, N-stearoyldihydrosphingosine, and N-behenoyldihydrosphingosine, N-docosanoyl-N-methyl-D-glucamine, cetyllic acid N-(2-hydroxyethyl)-N-(3-cetyloxy-2-hydroxypropyl)amide and bis(N-hydroxyethyl-N-cetyl)malonamide; and mixtures thereof. N-Oleoyldihydrosphingosine will preferably be used.

The solid fatty substances are preferably chosen from solid fatty acids, solid fatty alcohols, and mixtures thereof.

According to a preferred embodiment, the composition according to the invention comprises at least one liquid fatty substance, preferentially chosen from liquid hydrocarbons containing more than 16 carbon atoms, plant oils, liquid fatty alcohols and liquid fatty esters, silicone oils and mixtures thereof.

Preferentially, the liquid fatty substance(s) is (are) chosen from liquid hydrocarbons comprising more than 16 carbon atoms, in particular liquid petroleum jelly, liquid fatty alcohols, and mixtures thereof.

According to another preferred embodiment, the composition according to the invention comprises at least one solid fatty substance, preferentially chosen from solid fatty alcohols.

According to another preferred embodiment, the composition according to the invention comprises at least one liquid fatty substance and at least one solid fatty substance, preferentially at least one liquid hydrocarbon comprising more than 16 carbon atoms and at least one solid fatty alcohol.

When the composition according to the invention comprises one or more fatty substances, the total content of said

18

fatty substance(s) preferably ranges from 5% to 80% by weight, more preferentially from 8% to 70% by weight, and better still from 10% to 65% by weight, relative to the total weight of the composition.

In one particular embodiment, the composition according to the invention comprises one or more fatty substances, the total content of said fatty substance(s) preferably ranging from 30% to 80% by weight, more preferentially from 35% to 70% by weight, and better still from 40% to 65% by weight, relative to the total weight of the composition.

In another particular embodiment, the composition according to the invention comprises one or more liquid fatty substances, the total content of the liquid fatty substance(s) preferably ranging from 30% to 80% by weight, more preferentially from 35% to 70% by weight, and better still from 40% to 65% by weight, relative to the total weight of the composition.

Surfactants

The composition according to the present invention can comprise one or more surfactants. These may be chosen preferentially from anionic surfactants, non-ionic surfactants, cationic surfactants and/or mixtures thereof.

The term "anionic surfactant" is understood to mean a surfactant including, as ionic or ionizable groups, only anionic groups. These anionic groups are preferably chosen from the following groups: CO₂H, CO₂⁻, SO₃H, SO₃⁻, OSO₃H, OSO₃⁻, H₂PO₃, HPO₃, PO₃²⁻, H₂PO₂, HPO₂⁻, PO₂²⁻, POH and PO⁻.

As examples of anionic surfactants that may be used in the composition according to the invention, mention may be made of alkyl sulfates, alkyl ether sulfates, alkylamido ether sulfates, alkylaryl polyether sulfates, monoglyceride sulfates, alkylsulfonates, alkylamidesulfonates, alkylarylsulfonates, α-olefinsulfonates, paraffin sulfonates, alkyl sulfosuccinates, alkyl ether sulfosuccinates, alkylamide sulfosuccinates, alkyl sulfoacetates, acyl sarcosinates, acyl glutamates, alkyl sulfosuccinamates, acyl isethionates and N—(C₁-C₄)alkyl N-acyltaurates, salts of alkyl monoesters of polyglycoside-polycarboxylic acids, acyl lactylates, salts of D-galactoside uronic acids, salts of alkyl ether carboxylic acids, salts of alkylaryl ether carboxylic acids, salts of alkylamido ether carboxylic acids; and the corresponding non-salified forms of all these compounds; the alkyl and acyl groups of all these compounds (unless specified otherwise) generally comprising from 6 to 24 carbon atoms and the aryl group generally denoting a phenyl group.

These compounds may be oxyethylenated and then preferably comprise from 1 to 50 ethylene oxide units.

The salts of C₆-C₂₄ alkyl monoesters of polyglycoside-polycarboxylic acids may be chosen from C₆-C₂₄ alkyl polyglycoside-citrates, C₆-C₂₄ alkyl polyglycoside-tartrates and C₆-C₂₄ alkyl polyglycoside-sulfosuccinates.

When the anionic surfactant(s) is (are) in salt form, they may be chosen from alkali metal salts such as the sodium or potassium salt and preferably the sodium salt, ammonium salts, amine salts and in particular amino alcohol salts or alkaline-earth metal salts such as the magnesium salt.

Examples of amino alcohol salts that may notably be mentioned include monoethanolamine, diethanolamine and triethanolamine salts, monoisopropanolamine, diisopropanolamine or triisopropanolamine salts, 2-amino-2-methyl-1-propanol salts, 2-amino-2-methyl-1,3-propanediol salts and tris(hydroxymethyl)aminomethane salts.

Alkali metal or alkaline-earth metal salts and in particular the sodium or magnesium salts are preferably used.

The anionic surfactants optionally present may be mild anionic surfactants, i.e. anionic surfactants without a sulfate function.

As regards mild anionic surfactants, mention may be made in particular of the following compounds and salts thereof, and also mixtures thereof: polyoxyalkylenated alkyl ether carboxylic acids; polyoxyalkylenated alkylaryl ether carboxylic acids; polyoxyalkylenated alkylamido ether carboxylic acids, in particular those comprising 2 to 50 ethylene oxide groups; alkyl D-galactoside uronic acids; acyl sarcosines; acyl glutamates; and alkylpolyglycoside carboxylic esters.

Use may be made most particularly of polyoxyalkylenated alkyl ether carboxylic acids, for instance lauryl ether carboxylic acid (4.5 OE), sold, for example, under the name Akypo RLM 45 CA from Kao.

Among the anionic surfactants mentioned above, use is preferably made of the sulfated surfactants such as the alkyl sulfates or alkyl ether sulfates, and the acyl glutamates, more preferentially the alkyl sulfates.

The non-ionic surfactant(s) that may be used in the composition of the present invention are in particular described, for example, in the *Handbook of Surfactants* by M. R. Porter, published by Blackie & Son (Glasgow and London), 1991, pages 116-178.

As examples of non-ionic surfactants, mention may be made of the following compounds, alone or as a mixture: oxyalkylenated (C₈-C₂₄)alkylphenols;

saturated or unsaturated, linear or branched, oxyalkylenated or glycerolated C₈-C₄₀ alcohols, preferably comprising one or two fatty chains;

saturated or unsaturated, linear or branched, oxyalkylenated C₈ to C₃₀ fatty acid amides;

esters of saturated or unsaturated, linear or branched, C₈ to C₃₀ acids and of polyethylene glycols;

fatty acid esters of sucrose;

esters of saturated or unsaturated, linear or branched, C₈ to C₃₀ acids and of sorbitol which are preferably oxyethylenated;

C₈-C₃₀ fatty acid esters of sorbitan;

polyoxyethylenated C₈-C₃₀ fatty acid esters of sorbitan; (C₈-C₃₀)alkyl(poly)glucosides, (C₈-C₃₀)alkenyl(poly)glucosides, which are optionally oxyalkylenated (0 to 10 oxyalkylene units) and comprising from 1 to 15 glucose units, (C₈-C₃₀)alkyl(poly)glucoside esters;

saturated or unsaturated oxyethylenated plant oils;

condensates of ethylene oxide and/or of propylene oxide;

N—(C₈-C₃₀)alkylglucamine and N—(C₈-C₃₀)acylmethylglucamine derivatives; amine oxides.

They are notably chosen from alcohols, α-diols and (C₁-C₂₀)alkylphenols, these compounds being ethoxylated, propoxylated or glycerolated, and bearing at least one fatty chain comprising, for example, from 8 to 24 carbon atoms, preferably from 8 to 18 carbon atoms, the number of ethylene oxide or propylene oxide groups possibly ranging notably from 1 to 200, and the number of glycerol groups possibly ranging notably from 1 to 30.

Mention may also be made of condensates of ethylene oxide and of propylene oxide with fatty alcohols; ethoxylated fatty amides preferably having from 1 to 30 ethylene oxide units, polyglycerolated fatty amides comprising on average 1 to 5, and in particular 1.5 to 4, glycerol groups, fatty acid esters of sucrose, fatty acid esters of polyethylene glycol, oxyethylenated plant oils, N—(C₆-C₂₄ alkyl)glu-

camine derivatives, amine oxides such as (C₁₀-C₁₄)alkylamine oxides or N—(C₁₀-C₁₄ acyl)aminopropylmorpholine oxides.

The C₈-C₃₀ and preferably C₁₂-C₂₂ fatty acid esters (especially monoesters, diesters and triesters) of sorbitan may be chosen from: sorbitan caprylate; sorbitan cocoate; sorbitan isostearate; sorbitan laurate; sorbitan oleate; sorbitan palmitate; sorbitan stearate; sorbitan diisostearate; sorbitan dioleate; sorbitan distearate; sorbitan sesquicaprylate; sorbitan sesquiisostearate; sorbitan sesquioleate; sorbitan sesquisteate; sorbitan triisostearate; sorbitan trioleate; and sorbitan tristearate.

The polyoxyethylenated C₈-C₃₀ fatty acid esters (especially monoesters, diesters and triesters) of sorbitan are preferably chosen from the oxyethylenated C₈-C₃₀ fatty acid ester(s) of sorbitan having from 1 to 30 ethylene oxide units, preferably from 2 to 20 ethylene oxide units, more preferably from 2 to 10 ethylene oxide units.

Preferentially, the oxyethylenated C₈-C₃₀ fatty acid ester(s) of sorbitan is (are) chosen from the oxyethylenated C₁₂-C₁₈ fatty acid esters of sorbitan, in particular from the oxyethylenated esters of lauric acid, myristic acid, cetyl acid and stearic acid and of sorbitan.

Preferably, the oxyethylenated C₈-C₃₀ fatty acid ester(s) of sorbitan is (are) chosen from oxyethylenated sorbitan monolaurate (4 OE) (Polysorbate-21), oxyethylenated sorbitan monolaurate (20 OE) (Polysorbate-20), oxyethylenated sorbitan monopalmitate (20 OE) (Polysorbate-40), oxyethylenated sorbitan monostearate (20 OE) (Polysorbate-60), oxyethylenated sorbitan monostearate (4 OE) (Polysorbate-61), oxyethylenated sorbitan monooleate (20 OE) (Polysorbate-80), oxyethylenated sorbitan monooleate (5 OE) (Polysorbate-81), oxyethylenated sorbitan tristearate (20 OE) (Polysorbate-65), oxyethylenated sorbitan trioleate (20 OE) (Polysorbate-85).

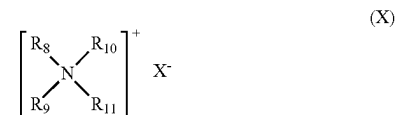
The non-ionic surfactant(s) is (are) preferably chosen from ethoxylated C₈-C₂₄ fatty alcohols comprising from 1 to 200 ethylene oxide groups, (C₆-C₂₄ alkyl)polyglycosides, the oxyethylenated C₈-C₃₀ fatty acid esters of sorbitan, and mixtures thereof.

The cationic surfactant(s) that may be used in the composition according to the invention is (are) generally chosen from optionally polyoxyalkylenated primary, secondary or tertiary fatty amines, quaternary ammonium salts, and mixtures thereof.

The fatty amines generally comprise at least one C₈-C₃₀ hydrocarbon-based chain. Among the fatty amines that may be used according to the invention, examples that may be mentioned include stearylamidopropyltrimethylamine and distearylamine.

Examples of quaternary ammonium salts that may notably be mentioned include:

those corresponding to the general formula (X) below:



in which the groups R₈ to R₁₁, which may be identical or different, represent a linear or branched aliphatic group comprising from 1 to 30 carbon atoms, or an aromatic group such as aryl or alkylaryl, at least one of the groups R₈ to R₁₁ comprising from 8 to 30 carbon atoms and preferably from

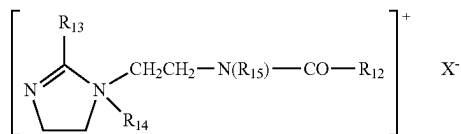
21

12 to 24 carbon atoms. The aliphatic groups may include heteroatoms such as especially oxygen, nitrogen, sulfur and halogens.

The aliphatic groups are chosen, for example, from C₁-C₃₀ alkyl, C₁-C₃₀ alkoxy, polyoxy(C₂-C₆)alkylene, C₁-C₃₀ alkylamide, (C₁₂-C₂₂)alkylamido(C₂-C₆)alkyl, (C₁₂-C₂₂)alkyl acetate and C₁-C₃₀ hydroxyalkyl groups; X⁻ is an anion chosen from the group of halides, phosphates, acetates, lactates, (C₁-C₄)alkyl sulfates and (C₁-C₄)alkylsulfonates or (C₁-C₄)alkylarylsulfonates.

Among the quaternary ammonium salts of formula (X), preference is given, firstly, to tetraalkylammonium chlorides, for instance dialkyldimethylammonium or alkyltrimethylammonium chlorides in which the alkyl group includes from about 12 to 22 carbon atoms, in particular behenyltrimethylammonium chloride, distearyldimethylammonium chloride, cetyltrimethylammonium chloride or benzyldimethylstearyl ammonium chloride, or else, secondly, to distearyl ethylhydroxyethylmethylammonium methosulfate, dipalmitoylethylhydroxyethylammonium methosulfate or distearyl ethylhydroxyethylammonium methosulfate, or else, finally, to palmitylamidopropyltrimethylammonium chloride or stearamidopropyl dimethyl(myristyl acetate) ammonium chloride, sold under the name Ceraphyl® 70 by the company Van Dyk.

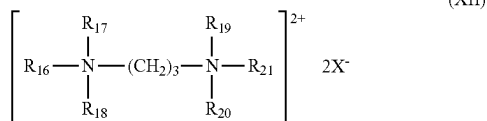
quaternary ammonium salts of imidazoline, for instance those of formula (XI) below:



in which R₁₂ represents an alkenyl or alkyl group including from 8 to 30 carbon atoms, for example tallow fatty acid derivatives, R₁₃ represents a hydrogen atom, a C₁-C₄ alkyl group or an alkenyl or alkyl group including from 8 to 30 carbon atoms, R₁₄ represents a C₁-C₄ alkyl group, R₁₅ represents a hydrogen atom or a C₁-C₄ alkyl group, and X⁻ is an anion chosen from the group of halides, phosphates, acetates, lactates, (C₁-C₄)alkyl sulfates and (C₁-C₄)alkylsulfonates or (C₁-C₄)alkylarylsulfonates.

Preferably, R₁₂ and R₁₃ denote a mixture of alkenyl or alkyl groups including from 12 to 21 carbon atoms, for example tallow fatty acid derivatives, R₁₄ denotes a methyl group, and R₁₅ denotes a hydrogen atom. Such a product is sold, for example, under the name Rewoquat® W 75 by the company Rewo.

quaternary diammonium or triammonium salts, in particular of formula (XII) below:



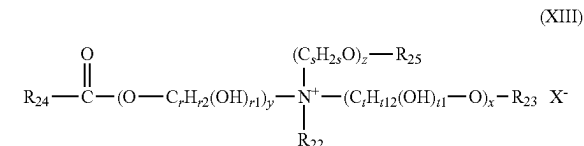
in which R₁₆ denotes an alkyl group containing approximately from 16 to 30 carbon atoms, which is optionally hydroxylated and/or interrupted with one or more oxygen atoms; R₁₇ is chosen from hydrogen, an alkyl group con-

22

taining from 1 to 4 carbon atoms or a group —(CH₂)₃—N⁺ (R16a)(R17a)(R18a), R16a, R17a, R18a, R₁₈, R₁₉, R₂₀ and R₂₁, which may be identical or different, are chosen from hydrogen or an alkyl group containing from 1 to 4 carbon atoms, and X⁻ is an anion chosen from the group of halides, acetates, phosphates, nitrates, (C₁-C₄)alkyl sulfates, (C₁-C₄)alkylsulfonates or (C₁-C₄)alkylarylsulfonates, in particular methyl sulfate and ethyl sulfate.

Such compounds are, for example, Finquat CT-P, sold by the company Finetex (Quaternium 89), and Finquat CT, sold by the company Finetex (Quaternium 75).

quaternary ammonium salts containing one or more ester functions, for instance those of formula (XIII) below:



in which: R₂₂ is chosen from C₁-C₆ alkyl groups and C₁-C₆ hydroxyalkyl or dihydroxyalkyl groups; R₂₃ is chosen from: the group-C(O)R₂₆, linear or branched, saturated or unsaturated C₁-C₂₂ hydrocarbon-based groups R₂₇, or a hydrogen atom; R₂₅ is chosen from: the group-C(O)R₂₈, linear or branched, saturated or unsaturated C₁-C₆ hydrocarbon-based groups R₂₉, or a hydrogen atom; R₂₄, R₂₆ and R₂₈, which may be identical or different, are chosen from linear or branched, saturated or unsaturated C₇-C₂₁ hydrocarbon-based groups; r, s and t, which may be identical or different, are integers from 2 to 6; r1 and t1, which may be identical or different, are 0 or 1; r2+r1=2 r and t1+t2=2 t, y is an integer from 1 to 10, x and z, which may be identical or different, are integers from 0 to 10, X⁻ is an organic or inorganic simple or complex anion, with the proviso that the sum x+y+z is from 1 to 15, that when x is 0 R₂₃ denotes R₂₇ and that when z is 0 R₂₅ denotes R₂₉.

The alkyl groups R₂₂ may be linear or branched, and more particularly linear.

Preferably, R₂₂ denotes a methyl, ethyl, hydroxyethyl or dihydroxypropyl group, and more particularly a methyl or ethyl group.

Advantageously, the sum x+y+z is from 1 to 10.

When R₂₃ is a hydrocarbon-based group R₂₇, it may be long and contain from 12 to 22 carbon atoms, or may be short and contain from 1 to 3 carbon atoms.

When R₂₅ is a hydrocarbon-based group R₂₉, it preferably contains 1 to 3 carbon atoms.

Advantageously, R₂₄, R₂₆ and R₂₈, which may be identical or different, are chosen from linear or branched, saturated or unsaturated C₁₁-C₂₁ hydrocarbon-based groups, and more particularly from linear or branched, saturated or unsaturated C₁₁-C₂₁ alkyl and alkenyl groups.

Preferably, x and z, which may be identical or different, are equal to 0 or 1.

Advantageously, y is equal to 1.

Preferably, r, s and t, which may be identical or different, are equal to 2 or 3, and even more particularly are equal to 2.

The anion X⁻ is preferably a halide, preferably chloride, bromide or iodide, a (C₁-C₄)alkyl sulfate, (C₁-C₄)alkylsulfonate or (C₁-C₄)alkylarylsulfonate. However, use may be made of methanesulfonate, phosphate, nitrate, tosylate, an anion derived from an organic acid, such as acetate

or lactate, or any other anion that is compatible with the ammonium bearing an ester function.

The anion X^- is even more particularly chloride, methyl sulfate or ethyl sulfate.

Use is made more particularly, in the composition according to the invention, of the ammonium salts of formula (XIII) in which: R_{22} denotes a methyl or ethyl group, x and y are equal to 1, z is equal to 0 or 1, r , s and t are equal to 2; R_{23} is chosen from: the group $-C(O)R_{26}$, methyl, ethyl or $C_{14}-C_{22}$ hydrocarbon-based groups, or a hydrogen atom, R_{25} is chosen from: the group $-C(O)R_{28}$, or a hydrogen atom, R_{24} , R_{26} and R_{28} , which may be identical or different, are chosen from linear or branched, saturated or unsaturated $C_{13}-C_{17}$ hydrocarbon-based groups, and preferably from linear or branched, saturated or unsaturated $C_{13}-C_{17}$ alkyl and alkenyl groups.

Advantageously, the hydrocarbon-based groups are linear.

Among the compounds of formula (XIII), examples that may be mentioned include the salts, especially the chloride or methyl sulfate, of diacyloxyethyltrimethylammonium, diacyloxyethylhydroxyethylmethylammonium, monoacyloxyethylhydroxyethylmethylammonium, triacyloxyethylmethylammonium or monoacyloxyethylhydroxyethyltrimethylammonium, and mixtures thereof. The acyl groups preferably contain 14 to 18 carbon atoms and are derived more particularly from a plant oil such as palm oil or sunflower oil. When the compound contains several acyl groups, these groups may be identical or different.

These products are obtained, for example, by direct esterification of triethanolamine, triisopropanolamine, an alkyl diethanolamine or an alkyl diisopropanolamine, which are optionally oxyalkylenated, with fatty acids or with mixtures of fatty acids of plant or animal origin, or by transesterification of the methyl esters thereof. This esterification is followed by quaternization by means of an alkylating agent, such as an alkyl halide, preferably methyl or ethyl halide, a dialkyl sulfate, preferably dimethyl or diethyl sulfate, methyl methanesulfonate, methyl para-toluenesulfonate, glycol chlorohydrin or glycerol chlorohydrin.

Such compounds are sold, for example, under the names Dehyquart® by the company Henkel, Stepanquat® by the company Stepan, Noxanium® by the company CECA or Rewoquat® WE 18 by the company Rewo-Witco.

The composition according to the invention may contain, for example, a mixture of quaternary ammonium monoester, diester and triester salts with a weight majority of diester salts.

Use may also be made of the ammonium salts containing at least one ester function that are described in U.S. Pat. Nos. 4,874,554 and 4,137,180.

Use may also be made of the behenoylhydroxypropyltrimethylammonium chloride sold, for example, by the company KAO under the name Quartamin BTC 131.

Preferably, the ammonium salts containing at least one ester function contain two ester functions.

Among the cationic surfactants, it is more particularly preferred to choose cetyltrimethylammonium, behenyltrimethylammonium and dipalmitoyl ethylhydroxyethylmethylammonium salts, and mixtures thereof, and more particularly behenyltrimethylammonium chloride, cetyltrimethylammonium chloride, and dipalmitoyl ethylhydroxyethylammonium methosulfate, and mixtures thereof.

Preferably, the surfactant(s) is (are) chosen from anionic surfactants, non-ionic surfactants, and mixtures thereof.

More preferentially, the surfactant(s) is (are) chosen from non-ionic surfactants, better still from ethoxylated C_8-C_{24} fatty alcohols comprising from 1 to 200 ethylene oxide

groups, (C_6-C_{24} alkyl)polyglycosides, the oxyethylenated C_8-C_{30} fatty acid esters of sorbitan, and mixtures thereof.

When the composition comprises one or more surfactant(s), the total content of surfactant(s) in the composition preferably ranges from 0.01% to 35% by weight, more preferentially from 0.1% to 30% by weight, better still from 0.5% to 25% by weight, even better still from 1% to 22% by weight, relative to the total weight of the composition.

When the composition comprises one or more non-ionic surfactant(s), the total content of non-ionic surfactant(s) in the composition preferably ranges from 0.01% to 35% by weight, more preferentially from 0.1% to 30% by weight, better still from 0.5% to 25% by weight, even better still from 1% to 22% by weight, relative to the total weight of the composition.

Thickening Polymer

The composition according to the present invention may comprise one or more thickening polymer(s).

For the purposes of the invention, the term "thickening" is understood to mean a compound capable, by its presence, of increasing the viscosity of the medium by at least 50 cps at 25° C. and at a shear rate of 1 s^{-1} . Preferably, the thickening compound, introduced at 1% by weight in water or in a 50/50 water/alcohol mixture, achieves a viscosity at 25° C. of greater than 100 cps at a shear rate of 1 s^{-1} . These viscosities can be measured using in particular viscometers or rheometers having cone-plate geometry.

The thickening polymers can be chosen from associative polymers and non-associative polymers.

For the purposes of the present invention, the term "associative polymers" means water-soluble polymers that are capable, in an aqueous medium, of reversibly associating with each other or with other molecules.

Their chemical structure comprises at least one hydrophilic region and at least one hydrophobic region characterized by at least one C_8-C_{30} fatty chain.

The associative polymers according to the invention may be of anionic, cationic, amphoteric or non-ionic type, preferably of non-ionic or cationic type.

Mention may in particular be made of the polymers sold under the names Pemulen TR1 or TR2 by the company Goodrich (INCI: Acrylates/C10-30 Alkyl Acrylate Crosspolymer), Salcare SC90 by the company Ciba, Aculyn 22, 28, 33, 44 or 46 by the company Röhm & Haas and Elfacos T210 and T212 by the company Akzo.

Preferably, the associative polymers can be chosen from cellulose-based polymers.

Preferably, the associative polymer(s) is (are) chosen from celluloses modified with groups comprising at least one fatty chain.

Preferentially, the associative polymer(s) may be chosen from hydroxyethylcelluloses modified with groups including at least one fatty chain, such as alkyl, arylalkyl or alkylaryl groups, or mixtures thereof, and in which the alkyl groups are preferably C_8-C_{22} , and hydroxyethylcelluloses modified with polyalkylene glycol alkyl phenol ether groups, and mixtures thereof, preferably cetylhydroxyethylcellulose.

Preferentially, the associative polymer(s) may be chosen from quaternized cellulose derivatives, preferably chosen from quaternized hydroxyethylcelluloses modified with groups comprising at least one fatty chain, such as linear or branched alkyl groups, linear or branched arylalkyl groups, or linear or branched alkylaryl groups, preferably linear or branched alkyl groups, these groups comprising at least 8 carbon atoms, notably from 8 to 30 carbon atoms, better still

from 10 to 24, or even from 10 to 14, carbon atoms; or mixtures thereof, and even better still polyquaternium-67.

The thickening polymers may also be chosen from non-associative polymers, and in particular non-ionic cellulose-based polymers (hydroxyethylcellulose, hydroxypropylcellulose, carboxymethylcellulose), guar gum and non-ionic derivatives thereof (hydroxypropylguar), gums of microbial origin (xanthan gum, scleroglucan gum), crosslinked or non-crosslinked homopolymers and copolymers based on acrylic acid, on methacrylic acid or on acrylamidopropane-sulfonic acid, and mixtures thereof.

Among the non-associative polymers, an anionic (meth)acrylic polymer may also be chosen, such as homopolymers or copolymers of (meth)acrylic acid. Mention may for example be made of the compounds having the INCI name Carbomer.

Preferably, the non-associative polymers are chosen from guar gums.

The guar gums that may be used according to the invention may be non-ionic or cationic.

According to the invention, use may be made of chemically modified or unmodified non-ionic guar gums.

The unmodified non-ionic guar gums are, for example, the products sold under the name Vidogum GH 175 by the company Unipectine and under the names Meypro-Guar 50 and Jaguar C by the company Rhodia Chimie.

The modified non-ionic guar gums that may be used according to the invention are preferably modified with C1-C6 hydroxyalkyl groups.

Among the hydroxyalkyl groups, mention may be made, by way of example, of hydroxymethyl, hydroxyethyl, hydroxypropyl and hydroxybutyl groups.

These guar gums are well known from the prior art and may be prepared, for example, by reacting corresponding alkene oxides, for instance propylene oxides, with the guar gum so as to obtain a guar gum modified with hydroxypropyl groups.

The degree of hydroxyalkylation, which corresponds to the number of alkylene oxide molecules consumed by the number of free hydroxyl functions present on the guar gum, preferably ranges from 0.4 to 1.2.

Such non-ionic guar gums optionally modified with hydroxyalkyl groups are sold, for example, under the trade names Jaguar HP8, Jaguar HP60 and Jaguar HP120, Jaguar DC 293 and Jaguar HP 105 by the company Meyhall, or under the name Galactasol 4H4FD2 by the company Aqualon.

Also suitable for use are non-ionic guar gums modified with hydroxyalkyl groups, more especially hydroxypropyl groups, modified with groups comprising at least one C₆-C₃₀ fatty chain. Examples of such compounds that may be mentioned include, inter alia, the product Esaflor HM 22® (C₂₂ alkyl chain) sold by the company Lamberti, and the products RE210-18® (C₁₄ alkyl chain) and RE205-1® (C₂₀ alkyl chain) sold by the company Rhône-Poulenc.

The cationic guar gums that may more particularly be used according to the invention are guar gums comprising trialkylammonium cationic groups. Preferably, 2% to 30% and more preferentially still 5% to 20% by number of the hydroxyl functions of these guar gums bear trialkylammonium cationic groups.

Among these trialkylammonium groups, mention may most particularly be made of the trimethylammonium and triethylammonium groups.

Even more preferentially, these groups represent from 5% to 20% by weight relative to the total weight of the modified guar gum.

According to the invention, a guar gum modified with 2,3-epoxypropyltrimethylammonium chloride is preferably used.

These guar gums modified with cationic groups are products already known per se and are, for example, described in patents U.S. Pat. Nos. 3,589,578 and 4,013,307. Such products are moreover notably sold under the trade names Jaguar C 13 S, Jaguar C 15 and Jaguar C 17 by the company Meyhall.

Use is preferably made of a non-ionic guar gum and, among these non-ionic guar gums, more particularly guar gums modified with hydroxyalkyl groups.

When the composition comprises one or more thickening polymers, the total content of the thickening polymer(s) preferably ranges from 0.01% to 20% by weight, more preferentially from 0.05% to 10% by weight, better still from 0.075% to 5% by weight, even better still from 0.1% to 3% by weight, relative to the total weight of the composition.

20 Alkaline Agent

The composition according to the present invention may comprise one or more mineral, organic or hybrid alkaline agent(s).

Preferably, the composition according to the present invention comprises one or more mineral, organic or hybrid alkaline agent(s).

For the purposes of the present invention, the terms "alkaline agent" and "basifying agent" are used interchangeably.

The mineral basifying agent(s) is (are) preferably chosen from aqueous ammonia, alkali metal carbonates or bicarbonates such as sodium (hydrogen)carbonate and potassium (hydrogen)carbonate, alkali metal or alkaline-earth metal phosphates such as sodium phosphates or potassium phosphates, sodium or potassium hydroxides, and mixtures thereof.

The organic basifying agent(s) is (are) preferably chosen from alkanolamines, amino acids, organic amines, oxyethylenated and/or oxypropylenated ethylenediamines, 1,3-diaminopropane, 1,3-diamino-2-propanol, spermine or spermidine and mixtures thereof.

The term "alkanolamine" is intended to mean an organic amine comprising a primary, secondary or tertiary amine function, and one or more linear or branched C₁-C₈ alkyl groups bearing one or more hydroxyl radicals.

Organic amines chosen from alkanolamines such as monoalkanolamines, dialkanolamines or trialkanolamines comprising one to three identical or different C₁-C₄ hydroxyalkyl radicals are in particular suitable for carrying out the invention.

In particular, the alkanolamine(s) is (are) chosen from monoethanolamine (MEA), diethanolamine, triethanolamine, monoisopropanolamine, diisopropanolamine, N,N-dimethylethanolamine, 2-amino-2-methyl-1-propanol, trisopropanolamine, 2-amino-2-methyl-1,3-propanediol, 3-amino-1,2-propanediol, 3-dimethylamino-1,2-propanediol, tris(hydroxymethyl)aminomethane and mixtures thereof.

Advantageously, the amino acids are basic amino acids comprising an additional amine function. Such basic amino acids are preferably chosen from histidine, lysine, arginine, ornithine and citrulline.

The organic amine may also be chosen from organic amines of heterocyclic type. Besides histidine that has already been mentioned in the amino acids, mention may in particular be made of pyridine, piperidine, imidazole, triazole, tetrazole and benzimidazole. The organic amine may

27

also be chosen from amino acid dipeptides. As amino acid dipeptides that may be used in the present invention, mention may notably be made of carnosine, anserine and balenine. The organic amine may also be chosen from compounds comprising a guanidine function. As amines of this type other than arginine that may be used in the present invention, mention may especially be made of creatine, creatinine, 1,1-dimethylguanidine, 1,1-diethylguanidine, glycocyamine, metformin, agmatine, n-amidoalanine, 3-guanidinopropionic acid, 4-guanidinobutyric acid and 2-([amino(imino)methyl]amino)ethane-1-sulfonic acid.

Use may be made in particular of guanidine carbonate or monoethanolamine hydrochloride as hybrid compounds.

The alkaline agent(s) that may be used according to the invention is (are) preferably chosen from alkanolamines such as monoethanolamine, diethanolamine, triethanolamine; aqueous ammonia, carbonates or bicarbonates such as sodium (hydrogen)carbonate and potassium (hydrogen)carbonate and mixtures thereof, more preferentially from aqueous ammonia and alkanolamines, better still from alkanolamines.

When the composition comprises at least one alkaline agent, the total content of the alkaline agent(s) preferably ranges from 0.1% to 40% by weight, more preferentially from 0.5% to 30% by weight, better still from 1% to 20% by weight, even better still from 2% to 15% by weight, relative to the total weight of the composition.

According to an embodiment, the pH of the composition comprising at least one alkaline agent, when it is aqueous, is between 8 and 13; preferably between 9 and 12.

The pH of the composition may be adjusted to the desired value by means of acidic or alkaline agent(s) commonly used in the dyeing of keratin fibres, such as those described hereinabove, or alternatively using buffer systems known to those skilled in the art.

Solvents

The composition according to the invention may also comprise at least one organic solvent.

Examples of organic solvents that may be mentioned include linear or branched C₂ to C₄ alkanols, such as ethanol and isopropanol; polyols and polyol ethers, for instance 2-butoxyethanol, glycerol, propylene glycol, dipropylene glycol, propylene glycol monomethyl ether, diethylene glycol monomethyl ether and monoethyl ether, and also aromatic alcohols or ethers, for instance benzyl alcohol or phenoxyethanol, and mixtures thereof.

The organic solvent(s) may be present in a total amount ranging from 0.01% to 30% by weight, preferably ranging from 2% to 25% by weight, relative to the total weight of the composition.

In addition, the composition according to the invention is preferably an aqueous composition. The composition preferably comprises water in an amount of greater than or equal to 5% by weight, preferably greater than or equal to 10% by weight, better still greater than or equal to 15% by weight, relative to the total weight of the composition.

Additives

The composition according to the invention may optionally comprise one or more additives, different from the compounds of the invention and among which mention may be made of cationic, anionic, non-ionic or amphoteric polymers or mixtures thereof, antidandruff agents, anti-seborrhoeic agents, agents for preventing hair loss and/or for promoting hair regrowth, vitamins and provitamins including panthenol, sunscreens, mineral or organic pigments, plasticizers, solubilizers, opacifiers or pearlescent agents, antioxidants, hydroxy acids, mineral thickening agents, fra-

28

grances, preservatives, sequestrants other than N, N-dicarboxymethyl glutamic acid and its salts.

Of course, those skilled in the art will take care to choose this or these optional additional compound(s) so that the advantageous properties intrinsically associated with the composition according to the invention are not, or not substantially, detrimentally affected by the envisioned addition(s).

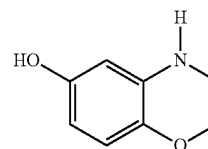
The above additives may generally be present in an amount, for each of them, of between 0 and 20% by weight, relative to the total weight of the composition.

Preferably, the composition according to the invention does not comprise chemical oxidizing agents.

According to a particular embodiment, the composition according to the invention comprises:

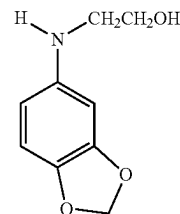
at least one oxidation coupler chosen from:

- i) 6-hydroxybenzomorpholine of formula (I), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof;



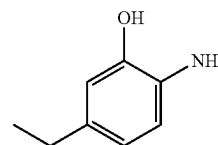
(I)

- ii) hydroxyethyl-3,4-methylenedioxyaniline of formula (II), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof;



(II)

- iii) 2-amino-5-ethylphenol of formula (III), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof, and



(III)

at least one compound chosen from N,N-dicarboxymethyl glutamic acid, one of its salts, and mixtures thereof;

at least one fatty substance,

at least one surfactant, and

at least one alkaline agent.

Process

The present invention also relates to a process for dyeing keratin fibres, preferably the hair, which comprises the step of applying to said keratin fibres an effective amount of a composition as described above.

29

The composition may be applied to wet or dry keratin fibres. On conclusion of the treatment, the keratin fibres are optionally rinsed with water, optionally washed with a shampoo and then rinsed with water, before being dried or left to dry.

Preferably, the process according to the invention comprises a step of mixing the composition according to the invention with an oxidizing composition comprising at least one chemical oxidizing agent. This mixing step is preferably performed at the moment of use, just before applying the composition resulting from the mixing to the hair.

More particularly, the chemical oxidizing agent(s) is (are) chosen from hydrogen peroxide, urea peroxide, alkali metal bromates or ferricyanides, peroxygenated salts, for instance persulfates, perborates, peracids and precursors thereof and percarbonates of alkali metals or alkaline-earth metals, and mixtures thereof. The chemical oxidizing agent is preferably hydrogen peroxide.

The oxidizing composition is preferably an aqueous composition. In particular, it comprises more than 5% by weight of water, preferably more than 10% by weight of water, and even more advantageously more than 20% by weight of water.

It may also comprise one or more organic solvents chosen from those listed previously; these solvents more particularly representing, when they are present, from 1% to 40% by weight and preferably from 5% to 30% by weight, relative to the weight of the oxidizing composition.

The oxidizing composition also preferably comprises one or more acidifying agents. Among the acidifying agents, examples that may be mentioned include mineral or organic acids, for instance hydrochloric acid, orthophosphoric acid, sulfuric acid, carboxylic acids, for instance acetic acid, tartaric acid, citric acid or lactic acid, and sulfonic acids.

The oxidizing composition may additionally comprise fatty substances such as those described hereinabove, preferably chosen from fatty alcohols, liquid hydrocarbons comprising more than 16 carbon atoms and mixtures thereof, surfactants and polymers.

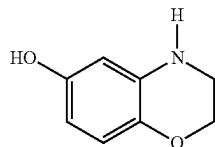
Usually, the pH of the oxidizing composition, when it is aqueous, is less than 7.

Preferably, the oxidizing composition comprises hydrogen peroxide as oxidizing agent, in aqueous solution, the concentration of which ranges, more particularly, from 0.1% to 50%, more particularly between 0.5% and 20% and even more preferentially between 1% and 15% by weight, relative to the weight of the oxidizing composition.

Preferably, at least one of the (dye or oxidizing) compositions is aqueous.

Preferably, the process according to the invention comprises a step of applying to the hair a composition resulting from the mixing, at the time of use, of at least two compositions:

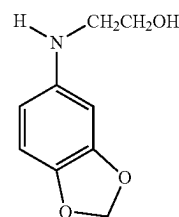
- a) a dyeing composition comprising:
 - at least one oxidation coupler chosen from:
 - i) 6-hydroxybenzomorpholine of formula (I), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof;



(I)

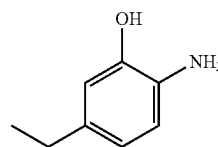
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- ii) hydroxyethyl-3,4-methylenedioxyaniline of formula (II), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof;



(II)

- iii) 2-amino-5-ethylphenol of formula (III), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof;



(III)

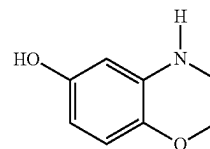
at least one compound chosen from N,N-dicarboxymethyl glutamic acid, one of its salts, and mixtures thereof;

and

- b) an oxidizing composition comprising one or more chemical oxidizing agents, preferably hydrogen peroxide.

According to one particular embodiment, the process according to the invention comprises a step of applying to the hair a composition resulting from the mixing, at the time of use, of at least two compositions:

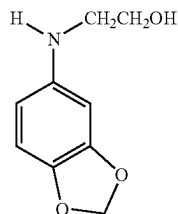
- a) a dyeing composition comprising:
 - at least one oxidation coupler chosen from:
 - i) 6-hydroxybenzomorpholine of formula (I), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof;



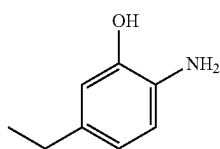
(I)

- ii) hydroxyethyl-3,4-methylenedioxyaniline of formula (II), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof;

31



iii) 2-amino-5-ethylphenol of formula (III), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof:



at least one compound chosen from N,N-dicarboxymethyl glutamic acid, one of its salts, and mixtures thereof;

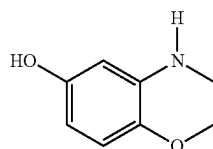
at least one fatty substance,
at least one surfactant, preferably non-ionic, and
at least one alkaline agent;

and

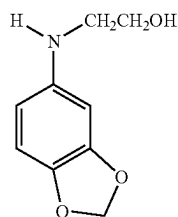
b) an oxidizing composition comprising one or more chemical oxidizing agents, preferably hydrogen peroxide.

The invention also relates to a composition comprising: at least one oxidation coupler chosen from:

i) 6-hydroxybenzomorpholine of formula (I), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof:



ii) hydroxyethyl-3,4-methylenedioxyaniline of formula (II), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof:



iii) 2-amino-5-ethylphenol of formula (III), one of its addition salts, its solvates and/or the solvates of its salts, and mixtures thereof:

(II)

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(III)

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(I)

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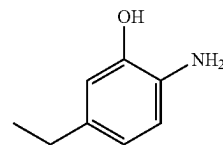
(II)

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(III)



at least one compound chosen from N, N-dicarboxymethyl glutamic acid, one of its salts, and mixtures thereof, and

one or more chemical oxidizing agents, preferably hydrogen peroxide, this composition being a ready-to-use composition.

This ready-to-use composition may comprise one or more ingredients among those described above.

Preferably, the pH of the ready-to-use composition is between 8 and 11, preferably between 9 and 10.7.

Kit

Another subject of the invention is a multi-compartment device for dyeing keratin fibres, comprising at least a first compartment containing the dyeing composition according to the invention and at least a second compartment containing an oxidizing composition as described above.

The compositions of the device according to the invention are packaged in separate compartments, optionally accompanied by suitable application means which may be identical or different, such as fine brushes, coarse brushes or sponges.

The device mentioned above may also be equipped with a means for dispensing the desired mixture onto the hair, for instance the devices described in patent FR 2586913.

Finally, the present invention relates to the use of a composition as described above for dyeing keratin fibres, and in particular the hair.

The following examples serve to illustrate the invention without, however, being limiting in nature.

EXAMPLES

In the examples that follow, all the amounts are given as weight percentage of active material (AM) relative to the total weight of the composition (unless stated otherwise).

Example 1

Dyeing Composition

The compositions A1, A2 and A3 according to the present invention and the comparative composition C were prepared using the ingredients of which the contents are indicated in the table below:

	A1 (invention)	A2 (invention)	A3 (invention)	C (comparative)
HYDROXYETHYL-3,4-METHYLENEDIOXYANILINE HCL	1.6 mmol	—	—	—
HYDROXYBENZOMORPHOLINE	—	1.6 mmol	—	—
2-AMINO-5-ETHYLPHENOL HCL	—	—	1.6 mmol	—
2-METHYL-5-HYDROXYETHYLAMINOPHENOL	—	—	—	1.6 mmol
m-AMINOPHENOL	0.065	0.065	0.065	0.065
2-METHYLRESORCINOL	0.33	0.33	0.33	0.33
2-METHOXYMETHYL-P-PHENYLENEDIAMINE	0.65	0.65	0.65	0.65
2-AMINO-3-HYDROXYPYRIDINE	0.025	0.025	0.025	0.025
2,4-DIAMINOPHENOXYETHANOL HCL	0.033	0.033	0.033	0.033
TETRASODIUM GLUTAMATE DIACETATE	0.095	0.095	0.095	0.095
LAURETH-12	7	7	7	7
DECETH-3	9	9	9	9
OLETH-30	4	4	4	4
PROPYLENE GLYCOL	10	10	10	10
MONOETHANOLAMINE	12.95	12.95	12.95	12.95
LAURIC ACID	3	3	3	3
GLYCOL DISTEARATE	2	2	2	2
SODIUM METABISULFITE	0.71	0.71	0.71	0.71
CARBOMER	0.39	0.39	0.39	0.39
TITANIUM DIOXIDE	0.15	0.15	0.15	0.15
POLYQUATERNIUM-22	1.52	1.52	1.52	1.52
CETEARYL ALCOHOL	11.5	11.5	11.5	11.5
HEXADIMETHRINE CHLORIDE	3	3	3	3
SILICA DIMETHYL Silylate	1.2	1.2	1.2	1.2
ASCORBIC ACID	0.25	0.25	0.25	0.25
Fragrance	Qs	Qs	Qs	Qs
WATER	Qs 100	Qs 100	Qs 100	Qs 100

Oxidizing Composition

The oxidizing composition B was prepared from the ingredients of which the contents are indicated in the table below.

TABLE 2

	B
HYDROGEN PEROXIDE	6
CETEARYL ALCOHOL	2.28
CETEARETH-25	0.57
TETRASODIUM ETIDRONATE	0.06
SODIUM SALICYLATE	0.035
GLYCEROL	0.5
TRIDECETH-2 CARBOXYAMIDE MEA	0.85
PHOSPHORIC ACID	Qs pH 2.2
Water	Qs 100

Dyeing Protocol 1

The dyeing compositions A1, A2, A3 and C are each mixed with the oxidizing composition B in a 1+1.5 weight ratio.

Each of the mixtures is applied to locks of sensitized hair (20% alkaline solubility) in a proportion of 5 g of mixture per 1 g of hair.

After a leave-on time of 30 minutes on a hot plate at 27° C., the hair is rinsed, washed with L'Oréal Professionnel Pro Classic universal concentrated shampoo, diluted to 10%, and dried.

Results

The colouring of the hair is evaluated in the L*a*b* system, using a Konica Minolta CM-3600A spectrophotometer (illuminant D65, angle 10°, specular component included) in the CIELab system.

In this system, L* represents the lightness. The lower the value of L*, the darker and more powerful the colouring

obtained. The chromaticity is measured by the values a* and b*, a* representing the red/green axis and b* the yellow/blue axis.

The results are shown in the table below:

TABLE 3

	L*
A1 + B (invention)	29.20
A2 + B (invention)	30.24
A3 + B (invention)	31.56
C + B (comparative)	33.96

The compositions A1, A2 and A3 according to the invention lead to a lower value of L*, and thus to better colour strength, compared to the comparative composition C.

Example 2

Dyeing Protocol 2

The dyeing compositions A1, A2, A3 and C are each mixed with the oxidizing composition B in a 1+1.5 weight ratio.

Each of the mixtures is applied to locks of hair containing 90% natural white (NW) hair and 90% natural white permanent-waved (PW) hair, in a proportion of 5 g of mixture per 1 g of hair.

After a leave-on time of 30 minutes on a hot plate at 27° C., the hair is rinsed, washed with L'Oréal Professionnel Pro Classic universal concentrated shampoo, diluted to 10%, and dried.

Results

The colouring of the hair is evaluated in the L*a*b* system, using a Konica Minolta CM-3600A spectrophotometer (illuminant D65, angle 10°, specular component included) in the CIELab system.

35

In this system, L^* represents the lightness. The lower the value of L^* , the darker and more powerful the colouring obtained. The chromaticity is measured by the values a^* and b^* , a^* representing the red/green axis and b^* the yellow/blue axis.

The selectivity is represented by the colour difference ΔE between the locks of dyed natural (NW) hair and dyed permanent-waved (PW) hair, ΔE being obtained from the formula:

$$\Delta E = \sqrt{(L^* - L_0^*)^2 + (a^* - a_0^*)^2 + (b^* - b_0^*)^2}$$

in which L^* represents the intensity and a^* and b^* represent the chromaticity of the dyed natural hair, and L_0^* represents the intensity and a_0^* and b_0^* represent the chromaticity of the dyed permanent-waved hair. The lower the value of ΔE , the lower the selectivity and the more uniform the colouration along the hair.

TABLE 4

	ΔE
A1 + B (invention)	6.48
A2 + B (invention)	6.52
A3 + B (invention)	5.91
C + B (comparative)	8.78

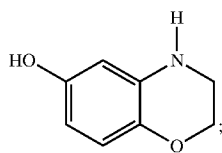
The compositions according to the invention lead to a lower value of ΔE , and thus to better selectivity, compared to the comparative composition.

The invention claimed is:

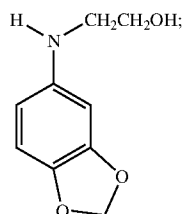
1. A composition comprising:

a) at least one oxidation coupler chosen from:

i) 6-hydroxybenzomorpholines of formula (I), addition salts thereof, solvates thereof, solvates of its salts thereof, or mixtures of two or more thereof:



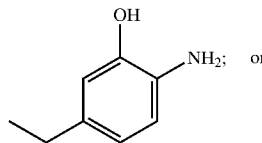
ii) hydroxyethyl-3,4-methylenedioxyanilines of formula (II), addition salts thereof, solvates thereof, solvates of its salts thereof, or mixtures of two or more thereof:



iii) 2-amino-5-ethylphenols of formula (III), addition salts thereof, solvates thereof, solvates of its salts thereof, or mixtures of two or more thereof:

36

(III)



iv) mixtures of two or more thereof; and
b) at least one compound chosen from N,N-dicarboxymethyl glutamic acid, salts thereof, or mixtures of two or more thereof.

2. The composition of claim 1, wherein the total amount of oxidation coupler(s) ranges from 0.001% to 20% by weight, relative to the total weight of the composition.

3. The composition of claim 1, wherein the total amount of oxidation coupler(s) ranges from 0.05% to 10% by weight, relative to the total weight of the composition.

4. The composition of claim 1, further comprising at least one oxidation base.

5. The composition of claim 4, wherein the at least one oxidation base is chosen from para-phenylenediamines, bis (phenyl) alkylenediamines, para-aminophenols, ortho-aminophenols, heterocyclic bases, addition salts thereof, or mixtures of two or more thereof.

6. The composition of claim 4, wherein the total amount of oxidation base(s) ranges from 0.001% to 20% by weight, relative to the weight of the composition.

7. The composition of claim 4, wherein the weight ratio of the total amount of oxidation base(s) to the total amount of oxidation coupler(s) ranges from 0.1:1 to 10:1.

8. The composition of claim 1, wherein the at least one compound is chosen from alkali metal salts of N,N-dicarboxymethyl glutamic acid.

9. The composition of claim 1, in which the total amount of compound(s) chosen from N,N-dicarboxymethyl glutamic acid, salts thereof, or mixtures thereof ranges from 0.001% to 15% by weight, relative to the total weight of the composition.

10. The composition of claim 1, further comprising at least one fatty substance.

11. The composition of claim 10, wherein the at least one fatty substance is chosen from liquid fatty substances, solid fatty substances, and mixtures thereof.

12. The composition of claim 11, wherein the at least one liquid fatty substance is chosen from liquid hydrocarbons comprising more than 16 carbon atoms, plant oils, liquid fatty alcohols, liquid fatty esters, silicone oils, or mixtures of two or more thereof.

13. The composition of claim 11, wherein the at least one solid fatty substance is chosen from solid fatty acids, solid fatty alcohols, solid esters of fatty acids, solid esters of fatty alcohols, waxes, ceramides, or mixtures of two or more thereof.

14. The composition of claim 1, further comprising at least one surfactant.

15. The composition of claim 14, wherein the at least one surfactant is chosen from anionic surfactants, non-ionic surfactants, or mixtures of two or more thereof.

16. The composition of claim 1, further comprising at least one alkaline agent.

17. The composition of claim 16, wherein the at least one alkaline agent is chosen from alkanolamines.

18. The composition of claim 1, further comprising at least one chemical oxidizing agent.

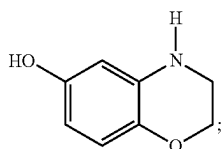
37

19. A method for dyeing keratin fibers comprising mixing a dyeing composition (A) with an oxidizing composition (B),

wherein the dyeing composition (A) comprises:

a) at least one oxidation coupler chosen from:

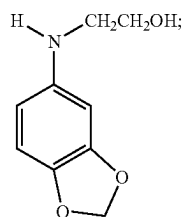
i) 6-hydroxybenzomorpholines of formula (I), addition salts thereof, solvates thereof, solvates of its salts thereof, or mixtures of two or more thereof;



(I)

15

ii) hydroxyethyl-3,4-methylenedioxyanilines of formula (II), addition salts thereof, solvates thereof, solvates of its salts thereof, or mixtures of two or more thereof;

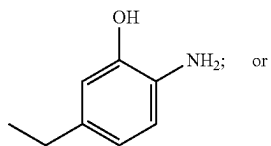


(II)

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iii) 2-amino-5-ethylphenols of formula (III), addition salts thereof, solvates thereof, solvates of its salts thereof, or mixtures of two or more thereof;



(III)

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iv) mixtures of two or more thereof; and

b) at least one compound chosen from N,N-dicarboxymethyl glutamic acid, salts thereof, or mixtures of two or more thereof; and

wherein the oxidizing composition (B) comprises at least one chemical oxidizing agent; and applying the mixture to the keratin fibers.

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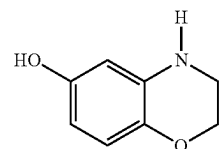
38

20. A kit comprising:

a first compartment comprising a dyeing composition comprising:

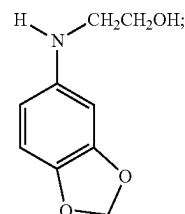
a) at least one oxidation coupler chosen from:

i) 6-hydroxybenzomorpholines of formula (I), addition salts thereof, solvates thereof, solvates of its salts thereof, or mixtures of two or more thereof;



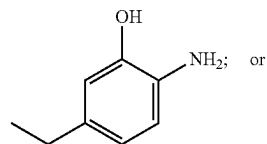
(I)

ii) hydroxyethyl-3,4-methylenedioxyanilines of formula (II), addition salts thereof, solvates thereof, solvates of its salts thereof, or mixtures of two or more thereof;



(II)

iii) 2-amino-5-ethylphenols of formula (III), addition salts thereof, solvates thereof, solvates of its salts thereof, or mixtures of two or more thereof;



(III)

iv) mixtures of two or more thereof; and

b) at least one compound chosen from N,N-dicarboxymethyl glutamic acid, salts thereof, or mixtures of two or more thereof; and

a second compartment comprising an oxidizing composition comprising at least one chemical oxidizing agent.

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